TOPICS IN GEOMETRIC REPRESENTATION THEORY

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Abstract. The goal of this course is to introduce many geometric objects and tools that are in the heart of GRT. Below is the tentative plan of the course:

- Part I. I will talk about the geometry of complex semisimple Lie algebras and groups. Specific topics include
  (1) The geometry of flag variety, nilpotent cone, Grothendieck-Springer resolution, Steinberg variety and an introduction to Springer theory
  (2) The wonderful compactification of semisimple groups and more general toroidal compactifications of reductive groups.

  We will follow some standard text/notes (and some additional references):
  - Chriss-Ginzburg, *Representation theory and complex geometry*, mostly chapter 3 and part of chapter 2
  - Evens-Jones, *On the wonderful compactification*
- Part 2. We will focus on a class of completely integrable systems, called the universal centralizers J_G (or under other names: Toda systems, regular centralizer group schemes, or bi-Whittaker reductions), associated with a semisimple (or reductive) group G. This class of completely integrable systems lies at the crossroads of many different subjects in differential geometry, GRT (including geometric Langlands), (homological and enumerative) mirror symmetry, and mathematical physics (e.g. the class of Coulomb branches). Selected topics include:
  (1) A partial log-compactification of J_G, by Balibanu (as an application of (2) in part 1)
  (2) Symplectic geometry of J_G and sketch of its homological mirror symmetry (based on some of my recent work)
  (3) Depending on interest, I will sketch the proof of a hyperKahler metric on it constructed from gauge theory (following Donaldson, Kronheimer and Bielawski).
- Prerequisite: Basic knowledge of differential topology and complex geometry is assumed. Familiarity with semisimple Lie algebras and algebraic groups, as well as their representations, will be helpful but is not strictly required. We will follow the approach of Chriss–Ginzburg in recalling and supplementing the necessary background as needed.

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1. Introduction

Representation theory studies some group G with some structure by considering representations $(V, \rho\colon G \to \mathrm{GL}(V))$ where V is a vector space and ρ is a group homomorphism with suitable properties. This seems like a purely algebraic question, but it is already very hard even when G is a finite group (in the nonabelian case). We may now bring in various geometric methods to construct representations of G:

(1) If G acts on an algebraic variety X, then there is an action of G on the space of functions on X;
(2) If G acts on an algebraic variety X, then we may consider a G-equivariant line bundle $\mathscr{L}$ on X (which is a line bundle on $[X/G]$), we may consider the cohomology groups $H^i(X, \mathscr{L})$ which are representations of G. If X is projective, then these cohomology groups are finite-dimensional;
(3) If G acts on X, then it also acts on the cohomology $H^\bullet(X)$ for any generalized cohomology theory;
(4) There are also more hidden and subtle symmetries, where instead of actual group actions, we consider correspondences acting on cohomology groups. We will see examples of this later when we discuss Springer theory. In addition, we can lift this to the level of derived categories and consider Fourier-Mukai transforms;

1.1. **Overview of the course.** The first part of the course will focus on the basics of geometric representation theory, largely focusing on Springer theory, the Borel-Weil-Bott theorem, and the geometry of the wonderful compactification $\overline{G}$.

The second part of the course will focus on a completely integrable system called the Toda system and its homological mirror symmetry. One motivation for this is the notion of categorification, for example going from the Hecke algebra to the Hecke category, which will turn out to be a category of constructible sheaves.

One origin of categorification in algebraic geometry is Grothendieck's function-sheaf correspondence, which we will adapt to our setting (in particular, we will

not need ℓ-adic sheaves). One particularly simple example of this is upgrading the integers $\mathbb{Z}$ to the derived category of vector spaces over a field k, where the original integer is recovered as the Euler characteristic.

For a more sophisticated example, we may consider a smooth manifold X with a stratification $\mathscr{S} = \{S_\alpha\}$ by locally closed submanifolds. We may consider the space $\mathsf{Fun}_{\mathscr{S}}(X)$ of constructible functions (taking values in integers) with respect to $\mathscr{S}$, which are locally constant on each stratum S_α. The natural categorification of this is the derived category $\mathsf{Sh}_{\mathscr{S}}(X)$ of constructible sheaves on X, where all cohomology sheaves are locally constant on each stratum S_α. The original constructible function is recovered by taking the Euler characteristic of the stalk at each point. In addition, we may recover the functions as

$$K_0(\mathsf{Sh}_{\mathscr{S}}(X)) \cong \mathsf{Fun}_{\mathscr{S}}(X).$$

1.2. **Homological mirror symmetry.** We will now discuss a connection to symplectic geometry. If we have a stratification $\mathscr{S}$, then we may consider the union

$$T^*_{\mathscr{S}}X := \bigsqcup_{S_\alpha \in \mathscr{S}} T^*_{S_\alpha}X$$

of conormal bundles to the strata and consider the microlocal sheaf theory. In particular, each constructible sheaf $\mathscr{F}$ has a microlocal stalk at each $(x, \xi) \in T^*_{\mathscr{S}}X$ as in Figure 1. For example, if we have a restriction map $\mathscr{F}(U) \to \mathscr{F}(V)$, then the microlocal stalk is

$$\mu_{(x,\xi)}(\mathscr{F}) = \mathrm{Cone}(\mathscr{F}(U) \to \mathscr{F}(V))[\cdots],$$

where the shift is not canonical. One upshot of this approach is that

$$K_0(\mathsf{Sh}_{\mathscr{S}}(X)) \cong \mathsf{Fun}_{\mathscr{S}}(X) \cong H^{\mathrm{BM}}_{\mathrm{top}}(T^*_{\mathscr{S}}X).$$

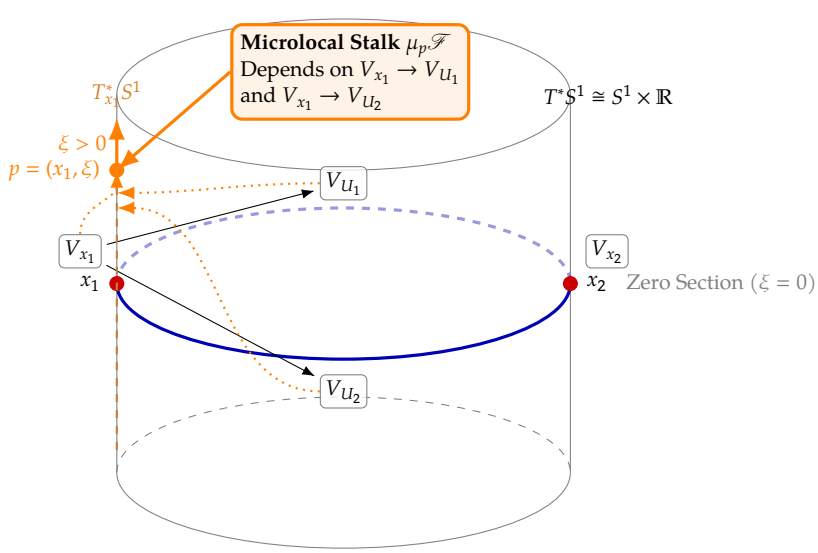


FIGURE 1. A microlocal stalk

The following theorem inspired many subsequent works.

Theorem 1.1 (Nadler-Zaslow)**.** *The dg-category* $\mathsf{Sh}_c(X)$ *of constructible sheaves on* X *equivalent the Fukaya category* $\mathsf{Fuk}(T^*X)$ *of the cotangent bundle* T^*X *of* X*.*

We will now discuss a geometric interpretation. We may consider the characteristic cycle $\mathrm{CC}(\mathscr{F})$ of a constructible sheaf, which is a conic Lagrangian cycle in T^*X. However, the objects of the Fukaya category are smooth Lagrangian submanifolds with the cochain complexes of morphisms being given by counting interesection points and holomorphic polygons. See Figure 2 for a visualization. It is usually very difficult to calculate the Fukaya category directly, but the theorem allows us to translate these questions into questions about constructible sheaves, which are more manageable.

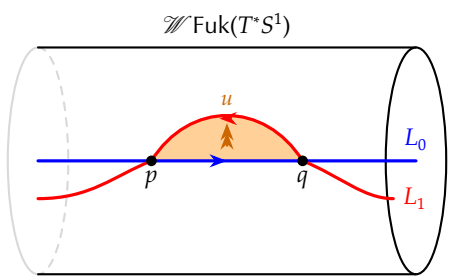


FIGURE 2. Two objects in the Fukaya category

Sketch of proof. We will begin by discussing why there is a fully faithful functor $\mathsf{Sh}_c(X) \to \mathsf{Fuk}(T^*X)$. The first step is to find some nice generators of the constructible sheaf category. For this, we consider open embeddings $j\colon U \hookrightarrow X$ and extension by zero $j_!\mathbb{C}_U$ of the constant sheaf on U. These have characteristic cycles given by the conormal bundle to the boundary ∂U (together with the zero section at U). We will send this sheaf $j_!\mathbb{C}_U$ to a smoothing of its characteristic cycle, as depicted in Figure 3.

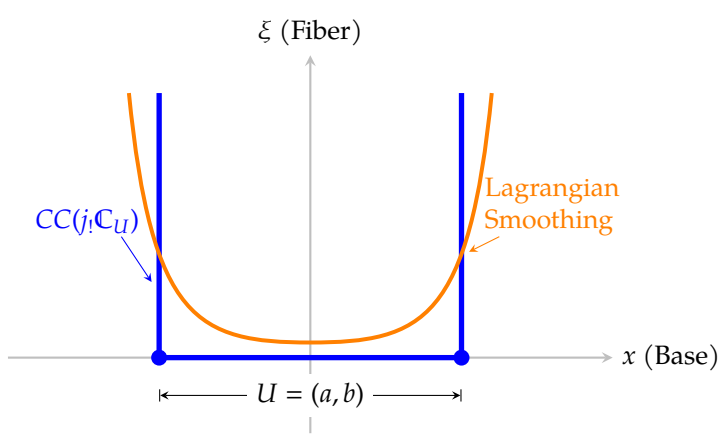


FIGURE 3. Characteristic cycle and its smoothing

To prove essential surjectivity, all of the Hom complexes in the Fukaya category can be calculated using Morse theory. For more details, see the original paper by Nadler-Zaslow. □

One reason to consider the Fukaya category is homological mirror symmetry, which was originally proposed by Kontsevich. For certain symplectic manifolds M, there is a *mirror partner* $\check{M}$, which is a complex algebraic variety, and Kontsevich conjectured that there is an equivalence of categories

$$\mathsf{Fuk}(M) \simeq \mathsf{Coh}(\check{M}).$$

The simplest example is when $M = T^*S^1$, where we need to consider the fully wrapped Fukaya category. We have an equivalence

$$\mathscr{W}\mathsf{Fuk}(T^*S^1) \simeq \mathsf{Loc}(S^1)$$

(modulo technical details), but the upshot is that all characteristic cycles are supported inside the zero section. Local systems are on S^1 are simply representations of $\pi_1(S^1) \cong \mathbb{Z}$, which are simply modules over the group algebra $\mathbb{C}[\mathbb{Z}] \cong \mathbb{C}[t, t^{-1}] = \mathscr{O}(\mathbb{C}^\times)$. Ignoring finiteness conditions, these are simply quasicoherent sheaves on $\mathbb{C}^\times$.

Remark 1.2. For the purposes of representation theory, we will treat $\pi_1(S^1) = \mathbb{X}_\bullet(S^1)$ as cocharacters, which will be identified with the characters $\mathbb{X}^\bullet(\mathbb{C}^\times)$ on the mirror side. This picture generalizes if we consider algebraic tori of any rank and is a version of Langlands duality in the abelian case.

Remark 1.3. Homological mirror symemtry is closely related to the geometric Langlands correspondence, where we send the Hitchin system for G to the character variety for the Langlands dual group $G^\vee$. There are some technical issues, but the Betti geometric Langlands correspondence is essentially homological mirror symmetry here.

One modification of this is to complexify the Fukaya side into T^*T, and in this case the (appropriate version of the) wrapped Fukaya category still reduces to local systems on T. Traditionally, we considered T^*T_c (the cotangent bundle of the compact form) because of the SYZ mirror symmetry philosophy, which predicts that a manifold M and its mirror $\check{M}$ have dual Lagrangian torus fibrations. In general, it is very hard to use SYZ to construct mirrors because there are usually singular fibers, which are hard to control.

However, one way to produce a mirror is to believe in homological mirror symmetry and then try to find a variety $\check{M}$ such that

$$\mathsf{Fuk}(M) \simeq \mathsf{Coh}(\check{M}).$$

If the fibration has a Lagrangian section, then each smooth fiber becomes a Lie group, so $\mathsf{Fuk}(M)$ is expected to pick up a "convolution" symmetric monoidal structure, which should correspond to the tensor product of coherent sheaves in some kind of non-linear Fourier transform.

In the second part of the course, we will generalize the duality $T^*T \leftrightarrow T^\vee$ to the universal centralizer (or Toda system) J_G, where G is a semisimple complex Lie group. This is related to representation theory for two reasons.

- J_G is also called the bi-Whittaker reduction and represents something like the generic part of the representation of G, namely the "commutative part" of the Hecke category;
- If we add a loop to G, then we obtain the affine Toda system;
- There is a partial (log) compactification $\bar{J}_G^{\log}$ over the base $\mathfrak{g}^* /\!/ G$ constucted by Balibanu whose fibers are a kind of regular Hessenberg variety (a generalization of Springer fibers). The central fiber is the Peterson variety Y_P, which is a closed subvariety of the flag variety G/B and is related to the quantum cohomology of $G^\vee/P^\vee$ via the isomorphism
$$QH^*(G^\vee/P^\vee) \cong \mathscr{O}(Y_P).$$

Part 1. Geometry of complex semisimple Lie algebras and groups

2. Review of semisimple Lie algebras/groups

Eventually, the groups we are interested in are $\mathrm{SL}_n(\mathbb{C})$, $\mathrm{SO}_n(\mathbb{C})$, and $\mathrm{Sp}_{2n}(\mathbb{C})$ with Lie algebras $\mathfrak{sl}_n(\mathbb{C})$, $\mathfrak{so}_n(\mathbb{C})$, and $\mathfrak{sp}_{2n}(\mathbb{C})$. However, along the way we will encounter groups like Borels and unipotent groups.

2.1. Generalities.

Every Lie group has both a left and right action of G on itself given by the multiplication map
$$m\colon G \times G \to G.$$
We may consider the tangent space $\mathfrak{g} = T_e G$ at the identity, and left translation of vector fields gives an isomorphism
$$L\colon G \times \mathfrak{g} \xrightarrow{\sim} TG \qquad (g, \xi) \mapsto L_{g*}\xi.$$
The right action gives another trivialization
$$R\colon G \times \mathfrak{g} \xrightarrow{\sim} TG \qquad (g, \xi) \mapsto R_{g^{-1}*}\xi.$$
These two are related by the diagram
$$\begin{tikzcd} G \times \mathfrak{g} \arrow[r, "L"] \arrow[d, "\iota\times(-1)"'] & TG \arrow[d, "T\iota"] \\ G \times \mathfrak{g} \arrow[r, "R"] & TG, \end{tikzcd}$$
where ι is the map sending an element of G to its inverse. If we fix $\xi \in \mathfrak{g}$, then applying L gives us a left-invariant vector field ξ^ℓ on G, and similarly applying R gives a right-invariant vector field ξ^r.

Lemma 2.1.

(1) *Left (resp. right) invariant vector fields on G are closed under the Lie bracket;*
(2) *The left and right-invariant vector fields induce the same Lie bracket on $\mathfrak{g}$.*

The main idea of the proof is that the Lie bracket is preserved by diffeomorphisms and in particular left (or right) translation.

Remark 2.2. If we have any action $a\colon G \times X \to X$, then taking the derivative
$$(\mathrm{d}a)_e\colon \mathfrak{g} \times X \to X$$

gives an infinitesimal action of G on X (as in a map from $\mathfrak{g}$ to vector fields on X). For example, L is simply

$$(\mathrm{d}m)_{e,\text{factor 2}} \colon G \times \mathfrak{g} \to TG$$

and a similar statement is true for R.

One important action of G on itself is the *adjoint action*, where G acts on itself by conjugation. On the tangent space, we have $(R_{g^{-1}})_* L_{g*} = L_{g*}(R_{g^{-1}})_*$. By definition, we now have

$$\mathrm{Ad}_g\, \xi := (R_{g^{-1}})_* L_{g*} \xi,$$

which induces a map

$$\mathrm{Ad} \colon G \to \mathrm{GL}(\mathfrak{g}).$$

Therefore, another equivalent definition of the Lie bracket is to simply differentiate the adjoint action. There is also an exponential map $\exp \colon \mathfrak{g} \to G$, and the Lie bracket is defined as

$$[\xi, \eta] := \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} \mathrm{Ad}_{\exp(t\xi)}\, \eta.$$

Remark 2.3. Any finite-dimensional abstract Lie algebra is isomorphic to a subalgebra of $\mathfrak{gl}_n$ by Ado's theorem, which says that every finite-dimensional Lie algebra has a finite-dimensional faithful representation. In particular, every finite-dimensional Lie group over $\mathbb{R}$ is locally an immersed subgroup of $\mathrm{GL}_n(\mathbb{R})$ for some n.

We will give some principles of Lie theory:

- The Lie group is (up to taking its universal cover) determined by its Lie algebra. Given a connected Lie group (H, e), then any discrete covering (H', e') inherits a unique group structure whose kernel lies in the center of H' (this works because the fundamental group of H is abelian due to the group structure). Conversely, any discrete central quotient of H is a Lie group with the same Lie algebra. If we work analytically, H has a universal cover, which is the unique simply connected Lie group with the same Lie algebra as H.
- If G and H are Lie groups such that G is connected, then any Lie group homomorphism
$$\rho \colon G \to H$$
is uniquely determined by its derivative $\mathrm{d}\rho \colon \mathfrak{g} \to \mathfrak{h}$, which is a Lie algebra homomorphism.
- For $\mathfrak{g} = \mathrm{Lie}\, G$ and any embedding $\mathfrak{h} \hookrightarrow \mathfrak{g}$, the subgroup of G generated by $\exp(\mathfrak{h})$ is an immersed Lie subgroup of G with Lie algebra $\mathfrak{h}$. For example, if $G = S^1 \times S^1$ and $\mathfrak{h}$ is a line with irrational slope, then H is immersed rather than embedded.
- Let G and H be finite dimensional Lie groups with G connected and simply connected. Then a map $f \colon T_e G \to T_e H$ is the differential of a Lie group homomorphism $\rho \colon G \to H$ if and only if it is a Lie algebra homomorphism. The standard trick to prove this is to consider the graph
$$\mathfrak{g} \hookrightarrow \mathfrak{g} \oplus \mathfrak{h} \qquad \xi \mapsto (\xi, f(\xi)),$$
which is a Lie algebra embedding. Then we get an immersed subgroup of $G \times H$ (this receives a map from G because G is simply connected), which we project down to H.

By Ado's theorem, we see that the embedding $\mathfrak{g} \hookrightarrow \mathfrak{gl}_n(\mathbb{R})$ gives an immersed subgroup $G \to \mathrm{GL}_n(\mathbb{R})$. In addition, the third item actually implies that there is a unique simply connected Lie group for any Lie algebra $\mathfrak{g}$.

Example 2.4. Let X be a projective variety. Then studying the identity component $\mathrm{Aut}(X)_0$ is similar to studying

$$T_e \mathrm{Aut}(X)_0 = H^0(X, TX),$$

which is a standard result in deformation theory. For example, if $X = G/P$ is a partial flag variety, then $H^0(X, TX) = \mathfrak{g}$ in almost all cases, but if $P = B$ (the full flag variety), then we really do have

$$H^0(X, TX) = \mathfrak{g}.$$

In particular, we have

$$\mathrm{Aut}(G/P)_0 = G/Z(G),$$

and when G is semisimple, the center is discrete.

We will call a Lie algebra homomorphism

$$\mathfrak{g} \to \mathfrak{gl}(V)$$

a *representation* of $\mathfrak{g}$. For example, we previously considered the adjoint representation

$$\mathrm{ad}\colon \mathfrak{g} \to \mathfrak{gl}(\mathfrak{g}).$$

We can also take the direct sums $V \oplus W$ and tensor products $V \otimes W$ of representations. It should be noted that the action on $V \otimes W$ should take the form

$$\xi(v \otimes w) = \xi v \otimes w + v \otimes \xi w$$

in analogy with differentiating the action of G on $V \otimes W$, which takes the form

$$g(v \otimes w) = gv \otimes gw.$$

2.2. **Rough classification of Lie algebras.** Given a Lie algebra $\mathfrak{g}$, there is a *lower central series*

$$\mathfrak{g} = D_0\mathfrak{g} \supset D_1(\mathfrak{g}) = [\mathfrak{g}, \mathfrak{g}] \supset \cdots \supset D_{k-1}(\mathfrak{g}) \supset D_k\mathfrak{g} = [\mathfrak{g}, D_{k-1}\mathfrak{g}] \cdots$$

and the smaller *derived series*

$$\mathfrak{g} = D^0\mathfrak{g} \supset D^1\mathfrak{g} = [\mathfrak{g}, \mathfrak{g}] \supset \cdots \supset D^{k-1}\mathfrak{g} \supset D^k\mathfrak{g} = [D^{k-1}\mathfrak{g}, D^{k-1}\mathfrak{g}].$$

Definition 2.5. A Lie algebra is *nilpotent* if the lower central series becomes zero at some point.

One example of a nilpotent Lie algbera is if we consider upper-triangular matrices with zeroes on the diagonal (usually denoted $\mathfrak{n}$).

Definition 2.6. A Lie algebra is *solvable* if the derived series eventually vanishes.

An example of a solvable Lie algebra is the algebra $\mathfrak{b}$ of all upper-triangular matrices. The reason for this is that $[\mathfrak{b}, \mathfrak{b}] = \mathfrak{n}$. If we write

$$\mathfrak{b} = \mathfrak{h} \oplus \mathfrak{n},$$

then we see that $[\mathfrak{h}, \mathfrak{h}] = 0$, $[\mathfrak{h}, \mathfrak{n}] = \mathfrak{n}$, and $[\mathfrak{n}, \mathfrak{n}] \subset \mathfrak{n}$.

Theorem 2.7 (Engel)**.** *Let $\mathfrak{g} \subset \mathfrak{gl}(V)$ be a Lie subalgebra such that every $X \in \mathfrak{g}$ is a nilpotent endomorphism of V. Then there exists a nonzero $v \in V$ such that $X(v) = 0$ for all $X \in \mathfrak{g}$.*

Engel's theorem implies that all $X \in \mathfrak{g}$ can be made (strictly) upper-triangular simultaneously.

Theorem 2.8 (Lie)**.** *Let $\mathfrak{g} \subset \mathfrak{gl}(V)$ be a solvable Lie subalgebra over $\mathbb{C}$. Then there is a common eigenvector for all $X \in \mathfrak{g}$.*

In particular, Lie's theorem implies that all $X \in \mathfrak{g}$ can be simultaneously made upper-triangular.

We will now consider all solvable ideals in $\mathfrak{g}$. In fact, if we have a finite collection of solvable ideals, then the sum is still solvable, so there exists a maximal solvable ideal of $\mathfrak{g}$, called the *radical* of $\mathfrak{g}$ and denoted $\mathrm{Rad}(\mathfrak{g})$. This fits into the exact sequence

$$0 \to \mathrm{Rad}(\mathfrak{g}) \to \mathfrak{g} \to \mathfrak{g}/\mathrm{Rad}(\mathfrak{g}) \to 0.$$

2.3. Semisimple and reductive Lie algebras.

Definition 2.9. A Lie algebra $\mathfrak{g}$ is *semisimple* if $\mathrm{Rad}(\mathfrak{g}) = 0$.

Example 2.10. The Lie algebra $\mathfrak{g}/\mathrm{Rad}(\mathfrak{g})$ is semisimple.

Definition 2.11. A Lie algebra is *reductive* if $\mathrm{Rad}(\mathfrak{g}) = Z(\mathfrak{g})$.

Remark 2.12. There are other characterizations of reductive Lie algebras, for example that the nilradical (maximal nilpotent ideal) $R_{\mathrm{nil}}(\mathfrak{g})$, which fits into the exact sequence

$$0 \to R_{\mathrm{nil}}(\mathfrak{g}) \to \mathfrak{g} \to \mathfrak{g}/R_{\mathrm{nil}}(\mathfrak{g}) \to 0,$$

vanishes.

We now give some important properties of semisimple Lie algebras.

Theorem 2.13. *Let $\mathfrak{g}$ be a semisimple Lie algebra. Then every finite-dimensional representation of $\mathfrak{g}$ decomposes uniquely as a direct sum of irreducible representations.*

Sketch of proof. There are two ways to do this, either algebraically (which is tedious) or by using Weyl's unitary trick. The key fact using the unitary trick is that every complex semisimple Lie algebra $\mathfrak{g}$ has a unique compact real form $\mathfrak{g}_c$ satisfying $\mathfrak{g}_c \otimes_{\mathbb{R}} \mathbb{C} = \mathfrak{g}$ (being a real form) and being the Lie algebra of a simply connected compact Lie group.

For example, the compact real form of $\mathfrak{sl}_n(\mathbb{C})$ is $\mathfrak{su}_n$ (traceles anti-Hermitian matrices), which corresponds to $\mathrm{SU}_n \subset \mathrm{SL}_n(\mathbb{C})$. The unitary trick is to restrict to the action of the compact form G_c on V and then try to find a G_c-invariant Hermitian metric on V, which will split $W \subset V$ for free. The way to find the inner product is to pick any Hermitian metric $\langle -, - \rangle$ and a Haar measure μ on G_c such that

$$\int_{G_c} \mathrm{d}\mu = 1,$$

and define a new inner product by

$$\langle v, w \rangle_{G_c} := \int_{G_c} \langle gv, gw \rangle \, \mathrm{d}\mu(g)$$

(which is the same trick we use for finite groups). Finally, the G_c-representations lift to G, so we are done. $\square$

Theorem 2.14. *In fact, a Lie algebra is semisimple if and only if all of its finite-dimensional representations are semisimple.*

Proof. Consider the adjoint representation $\mathrm{ad}\colon \mathfrak{g} \to \mathfrak{gl}(\mathfrak{g})$. Because it is semisimple, we have a decomposition

$$\mathfrak{g} = \mathrm{Rad}(\mathfrak{g}) \oplus \mathfrak{g}/\mathrm{Rad}(\mathfrak{g}).$$

In particular, we must have $[\mathrm{Rad}(\mathfrak{g}), \mathfrak{g}_{\mathrm{ss}}] = 0$, where $\mathfrak{g}_{\mathrm{ss}} = \mathfrak{g}/\mathrm{Rad}(\mathfrak{g})$.

If $\mathrm{Rad}(g) \neq 0$, then by Lie's theorem says that any representation must have an eigenvector, so all irreducible representations must be 1-dimensional. This implies that

$$\mathfrak{g} = \ell_1 \oplus \cdots \oplus \ell_k \oplus \mathfrak{g}_{\mathrm{ss}} = Z(\mathfrak{g}) \oplus \mathfrak{g}_{\mathrm{ss}}.$$

An abelian Lie algebra has indecomposable representations which are not semisimple, for example representing a line by matrices of the form $\left[\begin{smallmatrix} 0 & * \\ 0 & 0 \end{smallmatrix}\right]$, so the radical must have vanished to begin with. □

Remark 2.15. Levi's theorem actually states that $\mathfrak{g} = \mathrm{Rad}(\mathfrak{g}) \rtimes \mathfrak{g}_{\mathrm{ss}}$.

Theorem 2.16 (Preservation of Jordan decomposition)**.** *Let $\mathfrak{g}$ be semisimple. For any $X \in \mathfrak{g}$, there exist $X_s, X_n \in \mathfrak{g}$ such that $X_s + X_n = X$ and for any finite-dimensional representation*

$$\rho\colon \mathfrak{g} \to \mathfrak{gl}(V),$$

the Jordan normal form of $\rho(X)$ has $\rho(X)_s = \rho(X_s)$ and $\rho(X)_n = \rho(X_n)$.

Remark 2.17. If we consider the adjoint representation of $\mathfrak{g}$ (which is faithful), then in fact the condition on the Jordan decomposition in any representation actually implies that $X = X_s + X_n$.

Example 2.18. If we consider $\mathfrak{g} = \mathbb{C}$, then the representation

$$t \mapsto \begin{bmatrix} t & t \\ 0 & t \end{bmatrix}$$

has $\rho(X)_s$ and $\rho(X)_n$ not in the image of ρ in general.

Definition 2.19. If $\mathfrak{g}$ is semisimple, then we say $X \in \mathfrak{g}$ is *semisimple* if $X = X_s$ and *nilpotent* if $X = X_n$.

We will now consider reductive Lie algebras. In fact, by Levi's theorem and the fact that $[Z(\mathfrak{g}), \mathfrak{g}_{\mathrm{ss}}] = 0$, $\mathfrak{g}$ is reductive if and only if it is the product of a semisimple and an abelian Lie algebra, which is equivalent to $\mathfrak{g} = Z(\mathfrak{g}) \oplus [\mathfrak{g}, \mathfrak{g}]$.

Definition 2.20. A Lie group is *semisimple* if its Lie algebra is semisimple.

We will now consider all complex semisimple Lie groups, which are actually affine algebraic groups. By Borel's theorem, they are actually all linear, meaning closed subgroups of some GL_n. If $\mathfrak{g}$ is semisimple, then there is a simply connected form G_{sc} and a tower down to the *adjoint form* $G_{\mathrm{sc}} = G_{\mathrm{sr}}/Z(G_{\mathrm{sc}})$.

Example 2.21. For the Lie algebra $\mathfrak{sl}_n$, then the simply connected form is SL_n and the adjoint form is PGL_n.

Example 2.22. If we consider $\mathfrak{sp}_{2n}$, then the simply connected form is $\mathrm{Sp}_{2n}(\mathbb{C})$ and the adjoint form is $\mathrm{PSp}_{2n}(\mathbb{C})$.

Example 2.23. For SO_n, there is a form SO_n, which for $n \geq 3$ has fundamental group $\mathbb{Z}/2$, so the simply connected form is the universal cover Spin_n and the adjoint form is PSO_n.

Remark 2.24. We may also consider simple Lie algebras, which are semisimple Lie algebras with no nontrivial ideals. Then the classical simple Lie algebras are the ones we have already listed and are the only infinite families.

There is also the Cartan-Killing criterion of semisimple Lie algebras.

Definition 2.25. An *invariant* bilinear form on $\mathfrak{g}$ is given by

$$B\colon \mathfrak{g} \times \mathfrak{g} \to \mathbb{C}$$

satisfying the condition that

$$\mathrm{ad}_x(B(y,z)) := B([x,y],z) + B(y,[x,z]) = 0.$$

Definition 2.26. The *Killing form* is the form

$$B(x,y) = \mathrm{Tr}(\mathrm{ad}_x \circ \mathrm{ad}_y).$$

Because the trace has cyclic invariance and the trace is invariant under conjugation, this is invariant and symmetric.

Theorem 2.27 (Cartan-Killing criterion)**.** *A Lie algebra $\mathfrak{g}$ is semisimple if and only if the Killing form is nondegenerate.*

In particular, the means that the adjoint and coadjoint representations are isomorphic.

Remark 2.28. If $Z(\mathfrak{g}) \neq 0$, then $\mathrm{ad}_x = 0$ for all $x \in Z(\mathfrak{g})$, so $B(x,-) = 0$ for B the Killing form.

We now briefly pause to consider issues of analytic vs algebraic categories. If we are considering **semisimple** Lie groups over $\mathbb{C}$, then they are in fact also (linear) algebraic groups. However, if we leave the semisimple setting, then there are differences. For example, if we consider $\mathfrak{g} = \mathbb{C}$, then the algebraic options are $(\mathbb{C},+) = \mathbb{G}_a$, $(\mathbb{C}^\times,\times) = \mathbb{G}_m$, and compact 1-dimensional tori, namely elliptic curves. Usually, in algebraic groups theory, we will only consider affine algebraic groups.

Theorem 2.29 (Borel)**.** *Every affine algebraic group G is linear, and has a closed immersion to some* $\mathrm{GL}(V)$.

In the analytic world, there is the map

$$\exp\colon \mathbb{C} \to \mathbb{C}^\times,$$

which is not algebraic. In particular, $\mathbb{G}_a$ and $\mathbb{G}_m$ are totally different and have no nontrivial morphisms between them (and we have to consider their formal completions to recover the exponential map). In fact, the category $\mathrm{Rep}(\mathbb{G}_m)$ is semisimple (prove this using the unitary trick for S^1), while the same is not true for $\mathbb{G}_a$ (consider the representation by matrices of the form $\left[\begin{smallmatrix} 1 & * \\ 0 & 1 \end{smallmatrix}\right]$). However, any analytic Lie group over $\mathbb{C}$ not containing a compact torus is algebraic.

We now proceed to defining reductive groups. Recall that we essentially need to figure out what to do with the center $Z(\mathfrak{g})$ of a reductive Lie algebra $\mathfrak{g}$. In the algebraic group world, we will consider *tori*, which are groups isomorphic to products of copies of $\mathbb{G}_m$.

Definition 2.30. A *reductive* group G is a finite quotient of the product of an (affine) algebraic torus and a semisimple algebraic group.

In particular, we have

$$G = Z(G) \times^{Z(G^{\mathrm{der}})} G^{\mathrm{der}},$$

where $G^{\mathrm{der}} = [G, G]$ is semisimple and $Z(G)$ might be disconnected. For example, $Z(\mathrm{SL}_n) = \mu_n$, which is clearly not connected. Alternatively, there is a quotient morphism

$$G \to G/Z(G^{\mathrm{der}}) = Z(G)/Z(G^{\mathrm{der}}) \times G_{\mathrm{ad}},$$

where again $G_{\mathrm{ad}} = G/Z(G)$. Here, we see that $Z(G)/Z(G^{\mathrm{der}})$ is always connected. Using Langlands duality, we can produce a cover by something with connected center. A third alternative is to rewrite the product as

$$G = Z(G)_0 \times^{Z(G^{\mathrm{der}})^0} G^{\mathrm{der}},$$

where $Z(G^{\mathrm{der}})^0 = Z(G)_0 \cap G^{\mathrm{der}}$.

Example 2.31. In the case where $G = \mathrm{GL}_n$, we have $Z(\mathrm{GL}_n) = \mathbb{C}^\times$ and $(\mathrm{GL}_n)^{\mathrm{der}} = \mathrm{SL}_n$, so we have

$$\mathrm{GL}_n = \mathbb{C}^\times \times^{\mu_n} \mathrm{SL}_n.$$

Reductive **groups** are characterized by complete reducibility of their finite-dimensional representations (in characteristic zero). In addition, algebraic and analytic morphisms between reductive groups agree, so we no longer need to worry about the difference between algebraic and analytic categories. Representations of reductive groups also satisfy the preservation of Jordan decompositions, namely that for any $g \in G$, there exist $g_s, g_u \in G$ such that $g = g_s g_u = g_u g_s$ and for any finite-dimensional representation

$$\rho \colon G \to \mathrm{GL}(V),$$

the Jordan decomposition of $\rho(g)$ satisfies $\rho(g)_s = \rho(g_s)$ and $\rho(g)_u = \rho(g_u)$.

Lemma 2.32. *If $\mathfrak{g}$ is semisimple, then any invariant bilinear form is symmetric. In addition, any invariant non-degenerate bilinear form is a scalar multiple of the Killing form.*

Proof. Any invariant bilinear form B induces a map

$$\mathfrak{g} \to \mathfrak{g}^*$$

as morphisms of $G := G_{\mathrm{ad}}$-representations. Recall that $\mathfrak{g}$ decomposes into a product

$$\mathfrak{g} = \mathfrak{g}_1 \times \cdots \times \mathfrak{g}_k$$

of simple Lie algebras, so we have a morphism

$$\mathfrak{g}_1 \times \cdots \times \mathfrak{g}_k \to \mathfrak{g}_1^* \times \cdots \times \mathfrak{g}_k^*.$$

Using the Killing form, we have an isomorphism $\mathfrak{g}_i \cong \mathfrak{g}_i^*$, so by Schur's lemma, the map $\mathfrak{g}_i \to \mathfrak{g}_j^*$ is zero if $i \neq j$ and a scalar multiple of the identity if $i = j$. Therefore, any invariant bilinear form is a sum of scalar multiples of Killing forms on each simple factor, which are all symmetric. The second statement follows similarly. $\square$

2.4. **Cartan subalgebras.** We will begin by considering general $\mathfrak{g}$ and specialize to semisimple ones later.

Definition 2.33. If $\mathfrak{a} \subset \mathfrak{g}$ is a Lie subalgebra, then the *normalizer* of $\mathfrak{a}$ in $\mathfrak{g}$ is given by

$$\mathfrak{n}_{\mathfrak{g}}(\mathfrak{a}) = \{x \in \mathfrak{g} : [x, \mathfrak{a}] \subset \mathfrak{a}\}.$$

Definition 2.34. A subalgebra $\mathfrak{h} \subset \mathfrak{g}$ is a *Cartan subalgebra* if

(1) $\mathfrak{h}$ is nilpotent;
(2) $\mathfrak{h} = \mathfrak{n}_{\mathfrak{g}}(\mathfrak{h})$ (as in it is equal to its own normalizer).

We will see later that every Lie algebra has a Cartan subalgebra.

For any $x \in \mathfrak{g}$, let

$$P_x(T) = \det(T\,\mathrm{Id} - \mathrm{ad}_x) = \sum_{i=0}^{n} a_i(x)T^i$$

denote the characteristic polynomial of ad_x. If we choose coordinates $x_1, \dots, x_n$ on $\mathfrak{g}$, then each $a_i(x)$ is a homogeneous polynomial in $x_1, \dots, x_n$ of degree $n - i$.

Definition 2.35. The *rank* of $\mathfrak{g}$ is the least i such that $a_i(x)$ is not identically zero.

Example 2.36. If $\mathfrak{g} \cong \mathbb{C}^n$ is abelian, then $P_x(T) = T^n$ for all x, so $\mathfrak{g}$ has rank n.

Remark 2.37. Note that $i < n$ in the nonabelian case because $a_n(x) = 1$ is never zero and $i > 0$ because $x \in \ker(\mathrm{ad}_x)$ for all x and therefore $a_0(x) = 0$ for all x.

From now on, we will denote the rank of $\mathfrak{g}$ by r.

Definition 2.38. We say that $x \in \mathfrak{g}$ is *strongly regular* if $a_r(x) \neq 0$.

Remark 2.39. We will see another notion of regularity later, so we call this strongly regular to distinguish it.

We will denote the set of strongly regular elements by $\mathfrak{g}_{\mathrm{sreg}}$. The definition of a strongly regular element implies the following.

Proposition 2.40. *The set $\mathfrak{g}_{\mathrm{sreg}}$ is a connected open dense subset of $\mathfrak{g}$ (in the Zariski topology and hence also the analytic topology).*

Theorem 2.41. *If $x \in \mathfrak{g}_{\mathrm{sreg}}$, then the generalized eigenspace of 0 for ad_x, denoted $\mathfrak{g}_x^0$, is a Cartan subalgebra of $\mathfrak{g}$.*

Note that by definition, we have $\dim \mathfrak{g}_x^0 = r$, so the dimension of this Cartan equals the rank of $\mathfrak{g}$. In fact, we have already described all Cartan subalgebras.

Theorem 2.42. *Let $\mathfrak{g}$ be a Lie algebra over $\mathbb{C}$. Then all Cartan subalgebras of $\mathfrak{g}$ are of the form $\mathfrak{g}_x^0$ for some $x \in \mathfrak{g}_{\mathrm{sreg}}$.*

These Cartan subalgebras are all conjugate under the action of

$$G_{\mathrm{inn}} = \left\langle e^{\mathrm{ad}_x} \mid x \in \mathfrak{g} \right\rangle \hookrightarrow \mathrm{Aut}(\mathfrak{g}),$$

so in particular G_{inn} acts transitively on the set of Cartan subalgebras.

Remark 2.43. If G is reductive, then $G_{\mathrm{inn}} = G_{\mathrm{ad}} = G/Z(G)$.

2.5. **Root systems.** We will now specialize to the semisimple case, which is much more explicit.

Theorem 2.44. *Let $\mathfrak{h}$ be a Cartan subalgebra of a semisimple Lie algebra $\mathfrak{g}$. Then*

(1) *$\mathfrak{h}$ is abelian;*
(2) *$\mathfrak{h}$ is its own centralizer, i.e. $Z_{\mathfrak{g}}(\mathfrak{h}) = \mathfrak{h}$;*
(3) *Every element of $\mathfrak{h}$ is semisimple.*
(4) *The restriction of the Killing form to $\mathfrak{h}$ is nondegenerate.*

Example 2.45. If $\mathfrak{g} = \mathfrak{sl}_n$, then the Cartan subalgebras are precisely the conjugates of the diagonal matrices. This has dimension $n-1$ (given the constraint that the trace vanishes), so the rank of $\mathfrak{sl}_n$ is $n-1$.

For a semisimple Lie algebra, if $x \in \mathfrak{h} = \mathfrak{g}_x^0$, then in fact $\mathfrak{h} = \ker \mathrm{ad}_x$ for a general choice of x. The necessary condition on x is that it is *regular semisimple*, which we can take to simply mean that $\ker \mathrm{ad}_x$ is as small as possible. Because we can simultaneously diagonalize all ad_x for $x \in \mathfrak{h}$, we have a decomposition

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in \Phi} \mathfrak{g}_\alpha,$$

where

$$\mathfrak{g}_\alpha = \{x \in \mathfrak{g} : [h, x] = \alpha(h)x \text{ for all } h \in \mathfrak{h}\}$$

and Φ is the set of nonzero $\alpha \in \mathfrak{h}^*$ such that $\mathfrak{g}_\alpha \neq 0$.

Example 2.46. Returning to $\mathfrak{sl}_n$, if we let

$$\mathfrak{h} = \left\{ \begin{bmatrix} x_1 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & x_n \end{bmatrix} \right\},$$

then twe see that

$$\left[\begin{bmatrix} x_1 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & x_n \end{bmatrix}, E_{ij} \right] = (x_i - x_j)E_{ij}.$$

If we refer to the coordinates on $\mathbb{C}^n$ as $e_1, \dots, e_n$, then the roots are given by

$$\Phi = \left\{ \alpha_{ij} = e_i^* - e_j^* \mid i \neq j \right\},$$

where $\alpha_{ij}(\begin{bmatrix} x_1 & 0 & 0 \\ 0 & \dots & 0 \\ 0 & 0 & x_n \end{bmatrix}) = x_i - x_j$. We refer to the set Φ as the *roots* of $\mathfrak{g}$ with respect to $\mathfrak{h}$ and $\mathfrak{g}_\alpha$ as the *root space* corresponding to α.

Because the Killing form is nondegenerate when restricted to $\mathfrak{h}$, we may identify $e_i^* - e_j^* \in \mathfrak{h}^*$ with $e_i - e_j \in \mathfrak{h}$. In particular, we may consider Φ as a subset of $\mathfrak{h}$ itself.

Corollary 2.47.

(1) *$\mathfrak{h}$ is a maximal abelian subalgebra of $\mathfrak{g}$;*
(2) *Every strongly regular element is semisimple;*
(3) *G_{ad} acts transitively on the set of all Cartan subalgebras of $\mathfrak{g}$.*

Warning 2.48. Not all maximal abelian subalgebras are Cartan subalgebras, for example the nilpotent matrices in $\mathfrak{sl}_2$.

Lemma 2.49. *Assume that G is semisimple. Let $T = N_G(\mathfrak{h})_0$ (usually, it is not connected). Then T is a maximal torus in G with $T \cong (\mathbb{C}^\times)^r$.*

Sketch of proof. Choose a compact real form $\mathfrak{g}_c \subset \mathfrak{g}$. We now consider the set

$$\mathfrak{g}_{\text{sreg}} \cap \mathfrak{g}_c \neq \varnothing.$$

Now let $\xi_c \in \mathfrak{g}_c$ be strongly regular. Then $Z_{\mathfrak{g}_c}(\xi_c) = \mathfrak{h}_c$ is a Cartan subalgebra of $\mathfrak{g}_c$. Next, the normalizer

$$N_{G_c}(\mathfrak{h}_c)_0 = Z_{G_c}(\xi_c)_0 = T_c$$

is a maximal torus of G_c of real dimension r, so we can conclude the proof by complexifying everything. □

Example 2.50. In the case of $\mathfrak{sl}_n$, if ξ is strongly regular, then we see that $a_r(\xi) \neq 0$, and so $\dim \ker \operatorname{ad}_\xi = r = n-1$, so ξ must be semisimple. In particular, the root space decomposition is

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{i \neq j} \mathfrak{g}_{\alpha_{ij}}.$$

Lemma 2.51. *Choose a root decomposition*

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in \Phi} \mathfrak{g}_\alpha.$$

Then $\xi \in \mathfrak{h}$ is strongly regular if and only if there exists $\alpha \in \Phi$ such that $\alpha(\xi) \neq 0$ for all $\alpha \in \Phi$.

Proof. Note that $\xi \in \mathfrak{h}$ is strongly regular if and only if $\ker \operatorname{ad}_\xi = \mathfrak{h}$, which holds if and only if $\operatorname{ad}_\xi|_{\mathfrak{g}_\alpha}$ is nonzero for all $\alpha \in \Phi$, which holds if and only if $\alpha(\xi) \neq 0$ for all $\alpha \in \Phi$. □

Remark 2.52. From now on, we will refer to strongly regular elements as *regular semisimple* elements in the semisimple case and refer to the set of such elements as $\mathfrak{g}_{\text{rs}}$. We will also denote

$$\mathfrak{h}^{\text{reg}} := \mathfrak{g}_{\text{rs}} \cap \mathfrak{h},$$

which is given by the equation

$$\prod_{\alpha \in \Phi} \alpha \neq 0.$$

Example 2.53. The Lie algebra $\mathfrak{g} = \mathfrak{sl}_2$ is generated by the matrices

$$e = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \qquad f = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \qquad \text{and} \qquad h = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix},$$

with the relations

$$[h, e] = 2e, \qquad [h, f] = -2f, \qquad \text{and} \qquad [e, f] = h.$$

Then we see that the Cartan is $\mathfrak{h} = \mathbb{C}h$ and the root spaces are $\mathfrak{g}_\alpha = \mathbb{C}e$ and $\mathfrak{g}_{-\alpha} = \mathbb{C}f$, where $\alpha(h) = 2$. Therefore, the root system is $\Phi = \{\alpha, -\alpha\}$ and is shown in Figure 4. In this case, a choice of α is the same as choosing a set $\Phi^+ \subset \Phi$ of positive roots. In this case, α is also a simple root, but in general positive and simple roots are not the same.

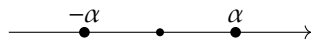


FIGURE 4. Root system of $\mathfrak{sl}_2$

Now set $\alpha^\vee = h \in \mathfrak{h}$, which is a *simple coroot*. Then we have $\langle \alpha^\vee, \alpha \rangle = 2$. We may now consider the maximal torus

$$\mathrm{SL}_2 \supset T = e^{2\pi i s h} = \left\{ \begin{bmatrix} t & \\ & t^{-1} \end{bmatrix} \right\} \cong \mathbb{C}/\mathbb{Z}h,$$

The adjoint action of T on $\mathfrak{g}$ is given by

$$\begin{aligned} \mathrm{Ad}\left(\begin{bmatrix} t & \\ & t^{-1} \end{bmatrix} \right) e &= \begin{bmatrix} t & \\ & t^{-1} \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} t^{-1} & \\ & t \end{bmatrix} \\ &= \begin{bmatrix} 0 & t \\ 0 & 0 \end{bmatrix} \begin{bmatrix} t^{-1} & \\ & t \end{bmatrix} \\ &= \begin{bmatrix} 0 & t^2 \\ 0 & 0 \end{bmatrix}. \end{aligned}$$

Note that $\alpha(h) = 2$, which matches the exponent. We may now consider the *cocharacter lattice*

$$\begin{aligned} \mathbb{X}_\bullet(T) &= \mathrm{Hom}(\mathbb{C}^\times, T) \\ &\cong \pi_1(T) \\ &\cong \mathbb{Z}h \\ &= \mathbb{Z}\alpha^\vee. \end{aligned}$$

If we project down to the adjoint form $\mathrm{PGL}_2 = \mathrm{SL}_2/\pm I$ we see in fact that

$$\mathbb{X}_\bullet(T_{\mathrm{ad}}) = \mathbb{Z} \cdot \left(\frac{1}{2} h \right).$$

Note that this cocharacter lattice is dual to the root lattice and is called the *coweight lattice*. We can now identify the *character lattice* of $T = T_{\mathrm{sc}}$ with the *weight lattice* $\mathbb{Z} \cdot \frac{1}{2}\alpha$, whereas $\mathbb{X}^\bullet(T_{\mathrm{ad}}) = \mathbb{Z} \cdot \alpha$ is the root lattice.

Now consider the exact sequence

$$1 \to Z(G_{\mathrm{sc}}) \to G_{\mathrm{sc}} \to G_{\mathrm{ad}} \to 1.$$

This induces the exact sequence

$$1 \to \pi_1(G_{\mathrm{sc}}) \to \pi_1(G_{\mathrm{ad}}) \to Z(G_{\mathrm{sc}}) \to 1.$$

In particular, we see once again that $\pi_1(G_{\mathrm{ad}}) \cong Z(G_{\mathrm{sc}})$. If we instead consider maximal tori, we have

$$1 \to Z(G_{\mathrm{sc}}) \to T_{\mathrm{sc}} \to T_{\mathrm{ad}} \to 1,$$

which induces the exact sequence

$$0 \to \mathbb{X}_\bullet(T_{\mathrm{sc}}) \to \mathbb{X}_\bullet(T_{\mathrm{ad}}) \to Z(G_{\mathrm{sc}}) \to 0$$

on fundamental groups. This gives us the identification

$$\pi_1(G_{\mathrm{ad}}) \cong \mathbb{X}_\bullet(T_{\mathrm{ad}}) \cong Z(G_{\mathrm{sc}}) \cong \mathbb{X}_\bullet(T_{\mathrm{ad}})/\mathbb{X}_\bullet(T_{\mathrm{sc}})$$

and relates the coroot and coweight lattices.

Example 2.54. In the case of $\mathfrak{sl}_3$, the root system is drawn in Figure 5. In particular, the roots are given by

$$\Phi = \left\{ \alpha_{ij} \colon \alpha_{ij} \begin{pmatrix} x_1 & & \\ & x_2 & \\ & & x_3 \end{pmatrix} = x_i - x_j \,\middle|\, i \neq j \right\}.$$

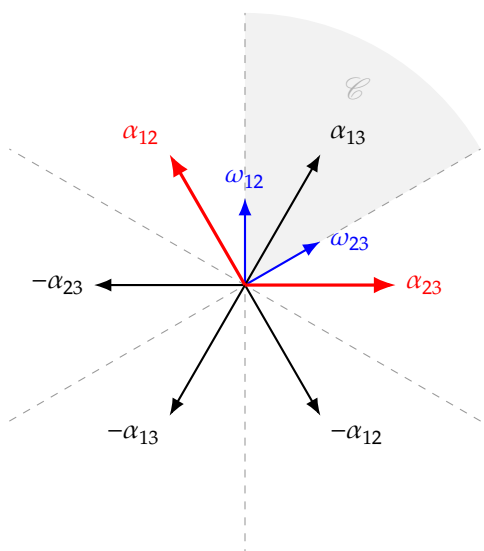


FIGURE 5. Root system of $\mathfrak{sl}_3$

In order to treat the coroots, note that

$$\mathfrak{sl}_2 \cong \mathfrak{g}_\alpha \oplus \mathfrak{g}_{-\alpha} \oplus [\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}]$$

for any root α. Therefore, we may choose $e_\alpha \in \mathfrak{g}_\alpha$ and $f_\alpha \in \mathfrak{g}_{-\alpha}$ such that

$$[e_\alpha, f_\alpha] = h_\alpha \in \mathfrak{h}.$$

For example, if we consider $\alpha = \alpha_{12}$, then we may choose

$$e_\alpha = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \qquad f_\alpha = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \qquad h_\alpha = \begin{bmatrix} 1 & & \\ & -1 & \\ & & 0 \end{bmatrix}.$$

We now have the coroot $\alpha_{ij}^\vee = h_{\alpha_{ij}}$ such that $\left\langle \alpha_{ij}^\vee, \alpha_{ij} \right\rangle = 2$.

To choose a basis, we cut $\mathfrak{h}^*$ by a general linear hyperplane and choose a positive direction. This gives a set Φ^+ of positive roots. For example, we will choose $\Phi^+ = \left\{ \alpha_{ij} \mid i < j \right\}$ for $\mathfrak{sl}_3$. Inside this, there is a **unique** set of simple roots $\Pi \subset \Phi^+$ satisfying

(1) It is a basis of $\mathfrak{h}_\mathbb{R}$;
(2) Any $\beta \in \Phi^+$ is a nonnegative integer combination of elements of Π.

For example, in this case, we have $\Pi = \{\alpha_{12}, \alpha_{23}\} \subset \Phi^+$. The simple coroots are then simply $\Pi^\vee = \left\{\alpha_{12}^\vee, \alpha_{23}^\vee\right\}$. Then the *Cartan matrix* is given by

$$C := \begin{bmatrix} \langle \alpha_{12}^\vee, \alpha_{12} \rangle & \langle \alpha_{12}^\vee, \alpha_{23} \rangle \\ \langle \alpha_{23}^\vee, \alpha_{12} \rangle & \langle \alpha_{23}^\vee, \alpha_{23} \rangle \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}.$$

We should also explain why the picture has the angles as shown. If we recall that the root lattice Q is identified with the character lattice $X^\bullet(T_{\mathrm{ad}})$, this is proved by considering the kernel $\ker Q \subset T_{\mathrm{ad}}$. In particular, anything killed by Q must actually act trivially on the root spaces and acts trivially on $\mathfrak{h}$ by definition, so it lives in the center of G_{ad}, which is trivial. Continuing as before, we consider the coweight lattice, which is spanned by the matrices

$$\omega_{i+1,i+2}^\vee := \frac{1}{3}(\varepsilon_i + \varepsilon_{i+1} - 2\varepsilon_{i+2}), \qquad \varepsilon_i = E_{ii}.$$

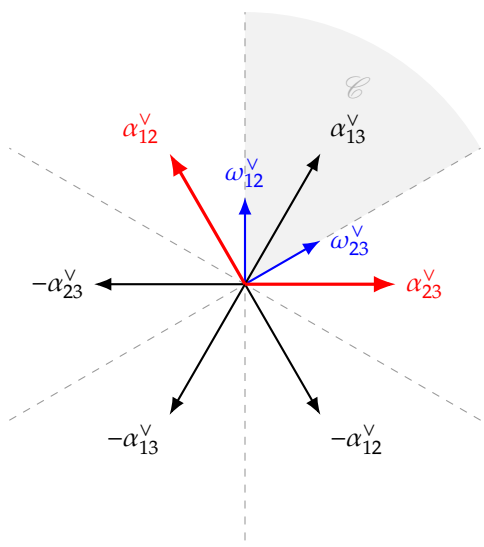


FIGURE 6. Coroot system of $\mathfrak{sl}_3$

Remark 2.55. The Cartan matrix is symmetric if and only if the Dynkin diagram is simply laced, i.e. has no multiple edges.

Unfortunately, we will not be able to discuss root systems in detail for time reasons. In type A (where $\mathfrak{g} = \mathfrak{sl}_{n+1}$), the picture generalizes that of $\mathfrak{sl}_3$, where we can choose $\Phi^+ = \left\{\alpha_{ij} \mid i < j\right\}$ and $\Pi = \left\{\alpha_{i,i+1} \mid 1 \le i \le n\right\}$. As a remark, we will also refer to the weight lattice as P and the coweight lattice as $P^\vee$. In general, we will interpret the fundamental weights ω_i as the dual basis to the simple coroots $\alpha_i^\vee$.

In the group picture, if we have an intermediate form G with maximal torus T, then we have the inclusions

$$\mathbb{X}_\bullet(T_{\mathrm{sc}}) \subset \mathbb{X}_\bullet(T) \subset \mathbb{X}_\bullet(T_{\mathrm{ad}})$$

induced by

$$T_{\mathrm{sc}} \to T \to T_{\mathrm{ad}}.$$

To find the center of G, we simply have the quotient $Z(G) = X_\bullet(T_{\mathrm{ad}})/X_\bullet(T)$, which comes from the long exact sequence on homotopy groups coming from

$$1 \to Z(G) \to T \to T_{\mathrm{ad}} \to 1.$$

To find the fundamental group of G, we apply this to the exact sequence

$$1 \to \pi_1(G) \to T_{\mathrm{sc}} \to T \to 1,$$

which gives us $\pi_1(G) = X_\bullet(T)/X_\bullet(T_{\mathrm{sc}})$. In particular, this specializes to

$$\mathbb{X}_\bullet(T_{\mathrm{ad}})/\mathbb{X}_\bullet(T_{\mathrm{sc}}) = Z(G_{\mathrm{sc}}) = \pi_1(G_{\mathrm{ad}}).$$

For example, if $G_{\mathrm{sc}} = \mathrm{SL}_3$, then we have

$$Z(G_{\mathrm{sc}}) = \mathbb{X}_\bullet(T_{\mathrm{ad}})/\mathbb{X}_\bullet(T_{\mathrm{sc}}) \cong \mathbb{Z}/3\mathbb{Z}.$$

Because this group has prime order, there are no intermediate forms between SL_3 and PGL_3.

Using the Killing form, there is the formula

$$\begin{aligned}\langle \alpha^\vee, \beta\rangle &= \frac{2(\alpha,\beta)}{(\alpha,\alpha)} \\ &= \frac{2\|\alpha\|\|\beta\|\cos\theta}{\|\alpha\|^2} \\ &= 2\frac{\|\beta\|}{\|\alpha\|}\cos\theta,\end{aligned}$$

and this actually implies that

$$\begin{aligned}\langle \alpha^\vee, \beta\rangle\langle \beta^\vee, \alpha\rangle &= 2\frac{\|\alpha\|}{\|\beta\|}\cos\theta \cdot 2\frac{\|\beta\|}{\|\alpha\|}\cos\theta \\ &= 4\cos^2\theta.\end{aligned}$$

Because this needs to be an integer, we see that for all roots, the angle θ between them must satisfy $4\cos^2\theta \in \mathbb{Z}$. This restricts the possible angles to $\frac{\pi}{2}, \frac{2\pi}{3}, \frac{\pi}{3}, \frac{\pi}{4}, \frac{3\pi}{4}, \frac{\pi}{6}$, or $\frac{5\pi}{6}$ unless $\alpha = \pm\beta$.

Remark 2.56. Root systems are classified by *Dynkin diagrams*, which is equivalent to classifying Cartan matrices. The entries of the Cartan matrix satisfy $C_{ii} = 2$ and $C_{ij} \le 0$. Non-symmetry can be explained by the fact that in some root systems, α_i and α_j may have different lengths. This gives arise to the notions of *long roots* and *short roots*. In fact, there are only two possible lengths for all roots (under the Weyl group action, any root can become simple).

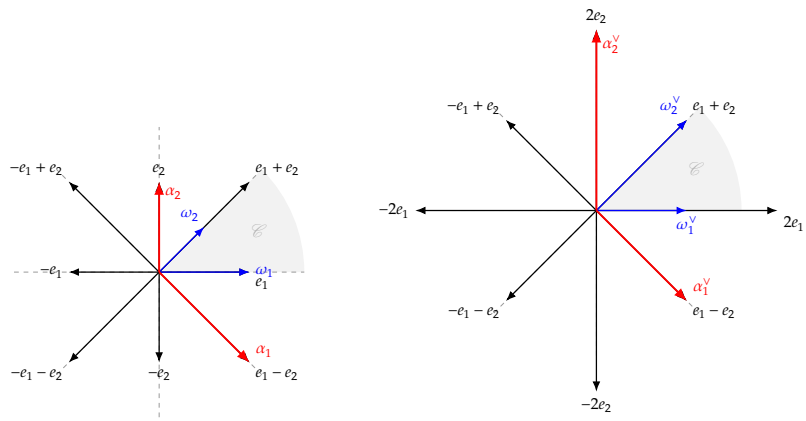
For an example of a root system with two lengths, consider the B_2 root system, whose root and coroot systems are shown in Figure 7.

2.6. **Weyl groups.** Recall that the maximal torus T has Lie algebra $\mathfrak{h}$. Then we define the *Weyl group*

$$W_T := N_G(T)/T.$$

Here, we know that $\mathfrak{n}_{\mathfrak{g}}(\mathfrak{h}) = \mathfrak{h}$, so $T \subset N_G(T)$.

Example 2.57. If $g = \mathfrak{sl}_n$, then $W_T \cong S_n$ because we can permute the diagonal entries.



(A) Root system of B_2 (B) Coroot system of B_2

FIGURE 7. Root and coroot systems of type B_2

The Weyl group acts on the set Φ of roots. Each root α corresponds to a reflection $s_\alpha \in W_T$ given by

$$\begin{aligned} s_\alpha(x) &= x - \langle \alpha^\vee, x\rangle \alpha \\ &= x - \frac{2(\alpha, x)}{(\alpha,\alpha)}\alpha. \end{aligned}$$

This satisfies $s_\alpha^2 = \mathrm{id}$ and fixes the hyperplane $\alpha^\perp \subset \mathfrak{h}^*$ pointwise.

In the $\mathfrak{sl}_n$ example, the root α_{ij} corresponds to the reflection that swaps the i-th and j-th coordinates. In particular, its kernel is the set of all matrices satisfying $x_i = x_j$.

The Weyl group is generated by the *simple reflections*, which correspond to the simple roots Π. We will refer to the simple roots as $\alpha_i \in \Pi$ and the corresponding reflections as $s_i = s_{\alpha_i}$. Then, the Weyl group has the presentation

$$W_T = \left\langle s_i \mid s_i^2 = e, \dots \right\rangle.$$

and is a finite *reflection group*.

The reflection hyperplanes $\alpha^\perp$ cut the space $\mathfrak{h}^*_{\mathbb{R}}$ into a set of chambers, which are fundamental domains for the Weyl action and are in bijection with the choices of positive roots Φ^+. In particular, the set Φ^+ corresponds to the chamber

$$\begin{aligned} C_{\Phi^+} &= \left\{\mu \in \mathfrak{h}^*_{\mathbb{R}} \mid \langle \alpha^\vee, \mu\rangle \geq 0 \text{ for all } \alpha^\vee \in \Phi^{\vee,+}\right\} \\ &= \left\{\mu \in \mathfrak{h}_{\mathbb{R}} \mid \langle \alpha^\vee, \mu\rangle \geq 0 \text{ for all } \alpha^\vee \in \Pi^{\vee,+}\right\} \\ &= \mathrm{Span}_{\mathbb{R}_{\geq 0}}\{\omega_i \mid 1 \leq i \leq r\}. \end{aligned}$$

Note also that the choice of a set of positive roots corresponds to a choice of Borel $\mathfrak{b} \supset \mathfrak{h}$. In particular, there is the direct sum decomposition

$$\mathfrak{b} = \mathfrak{h} \oplus \bigoplus_{\alpha \in \Phi^+} \mathfrak{g}_\alpha.$$

In addition, for any $\alpha, \beta \in \Phi$ we have

$$[g_\alpha, g_\beta] = \begin{cases} g_{\alpha+\beta} & \alpha+\beta \in \Phi, \\ \Bbbk\alpha^\vee & \alpha+\beta = 0, \\ 0 & \text{otherwise.} \end{cases}$$

2.7. **Regular elements.** Let $\xi \in \mathfrak{h}^{\mathrm{reg}} := \mathfrak{h} \smallsetminus \bigcup_{\alpha\in\Phi^+} \ker\alpha$. Note that W_T acts freely on $\mathfrak{h}^{\mathrm{reg}}$. Finally, let $r = \operatorname{rank}\mathfrak{g} = \dim\mathfrak{h}$.

Proposition 2.58. *Let $x \in \mathfrak{g}$. Then the centralizaer $Z_{\mathfrak{g}}(x)$ contains an r-dimensional abelian subalgebra.*

Proof. Since the regular semisimple locus in dense, there exists a sequence $x_n \to x$ of regular semisimple elements. For each x_n, its centralizer $\mathfrak{h}_n = Z_g(x_n)$ is r-dimensional ane abelian.

Finally, the compactness of the Grassmannian $\mathrm{Gr}(r, \mathfrak{g})$ implies that there is some subsequence $\mathfrak{h}_n = Z_{\mathfrak{g}}(x_n)$ which converges to some limit $\mathfrak{u}$. We need to check that $\mathfrak{u} \subset Z_{\mathfrak{g}}(x)$, but we know that if $y \in \mathfrak{u}$, there is some sequence $y_n \to y$ such that $y_n \in \mathfrak{h}_n$. Because the Lie bracket is continuous on both arguments, we see that

$$\begin{aligned} [x, y] &= \left[\lim_{n\to\infty} x_n, \lim_{n\to\infty} y_n\right] \\ &= \lim_{n\to\infty} [x_n, y_n] \\ &= 0. \end{aligned}$$

We now need to check that $\mathfrak{u}$ is abelian. But this follows similarly: if $y, z \in \mathfrak{u}$, there are sequences $y_n \to y$ and $z_n \to z$ such that $y_n, z_n \in \mathfrak{h}_n$. Then

$$\begin{aligned} [y, z] &= \left[\lim_{n\to\infty} y_n, \lim_{n\to\infty} z_n\right] \\ &= \lim_{n\to\infty} [y_n, z_n] \\ &= 0. \end{aligned}$$

□

Corollary 2.59. *For all $x \in \mathfrak{g}$, then $\dim Z_{\mathfrak{g}}(x) \geq r$ and $\dim Z_G(x) \geq r$. In addition, every maximal abelian subalgebra has dimension at least r.*

Definition 2.60. An element $x \in \mathfrak{g}$ is called *regular* if $\dim Z_{\mathfrak{g}}(x) = r$. Equivalently, the adjoint orbit $G \cdot x$ has maximal dimension.

Example 2.61. An example of a regular element which is not semisimple is

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \in \mathfrak{sl}_3.$$

Its centralizer is given by

$$Z_{\mathfrak{sl}_3}(x) = \left\{ \begin{bmatrix} 0 & a & b \\ 0 & 0 & a \\ 0 & 0 & 0 \end{bmatrix} \right\}$$

and is 2-dimensional.

We will now discuss a general criterion for regularity. Recall the Jordan decomposition

$$\mathfrak{g} \ni x = x_s + x_n, \qquad [x_s, x_n] = 0,$$

where x_s is semisimple and x_n is nilpotent. Because $\operatorname{ad} x = \operatorname{ad} x_s + \operatorname{ad} x_n = (\operatorname{ad} x)_s + (\operatorname{ad} x)_n$, we have

$$\begin{aligned} Z_{\mathfrak{g}}(x) &= \ker \operatorname{ad} x \\ &= \ker \operatorname{ad} x_s \cap \ker \operatorname{ad} x_n \\ &= Z_{\mathfrak{g}}(x_s) \cap Z_{\mathfrak{g}}(x_n). \end{aligned}$$

Proposition 2.62. *The element $x \in \mathfrak{g}$ is regular if and only if x_n is regular in the derived algebra of $\ell_{x_s} := Z_{\mathfrak{g}}(x_s)$.*

Remark 2.63. The centralizer ℓ_{x_s} is a reductive subalgebra of $\mathfrak{g}$, called a *Levi subalgebra*. Here, if $x_s \in \mathfrak{h}$ for some Cartan $\mathfrak{h}$ and

$$\mathfrak{h} = \mathfrak{h} \oplus \bigoplus_{\alpha \in \Phi} \mathfrak{g}_\alpha$$

is the root decomposition, then we have

$$Z_{\mathfrak{g}}(x_s) = \mathfrak{h} \oplus \bigoplus_{\substack{\alpha \in \Phi \\ \alpha(x_s)=0}} \mathfrak{g}_\alpha.$$

We now consider the root system

$$\mathfrak{h}^* \supset \ker x_s \supset \Phi' := \{\alpha \mid \alpha(x_s) = 0\} \subset \Phi.$$

This is the root system for $\ell_{x_s}^{\mathrm{der}}$ (note we need to take the derived subalgebra to eliminate the center). Finally, we note that

$$x_s \in Z(\ell_{x_s}) = \ker \Phi' \subset \mathfrak{h},$$

so the rank of the root system increases as we move to smaller strata of $\mathfrak{h}$ (with respect to the stratification by root hyperplanes). The structure of the Levi subalgebra is in fact

$$\ell_{x_s} = Z(\ell_{x_s}) \oplus \ell_{x_s}^{\mathrm{der}},$$

where the center is $(\Phi')^\perp$ and the semisimple part has root system Φ'.

Example 2.64. In $\mathfrak{sl}_3$, the matrix

$$\begin{bmatrix} a & & \\ & a & \\ & & -2a \end{bmatrix}$$

lies in the root hyperplane $\alpha_{12} = 0$ and its Levi subalgebra is given by traceless block-diagonal matrices of the form

$$\begin{bmatrix} * & * & 0 \\ * & * & 0 \\ 0 & 0 & * \end{bmatrix}.$$

Taking the derived subalgebra, we obtain a copy of $\mathfrak{sl}_2$ in the upper-left block. The center is simply matrices of the form

$$\begin{bmatrix} b & & \\ & b & \\ & & -2b \end{bmatrix}.$$

In $\mathfrak{sl}_n$, we can group the blocks by shared eigenvalues and proceed similarly. If the blocks have sizes $m_1, \dots, m_k$, then the derived subalgebra of the Levi is given by $\mathfrak{sl}_{m_1} \times \cdots \times \mathfrak{sl}_{m_k}$ and the center has dimension $k-1$.

Proof of Proposition 2.62. Note first that

$$\ell = \mathbb{Z}_{\mathfrak{g}}(x_s) = Z(\ell) \oplus \ell^{\mathrm{der}}.$$

Set-theoretically, we have

$$\begin{aligned}\mathbb{Z}_{\mathfrak{g}}(x) &= \mathbb{Z}_{\mathfrak{g}}(x_s) \cap \mathbb{Z}_{\mathfrak{g}}(x_n) \\ &= Z_{\ell}(x_n) \\ &= Z(\ell) \oplus Z_{\ell^{\mathrm{der}}}(x_n).\end{aligned}$$

If $\dim Z(\ell) = k$, then $\operatorname{rank} \ell^{\mathrm{der}} = \dim Z(\ell)^{\perp} = r - k$, where the orthogonal complement is taken in $\mathfrak{h}^*$. Therefore, we know that x is regular if and only if $\dim Z_{\ell^{\mathrm{der}}}(x_n) = r - k = \operatorname{rank} \ell^{\mathrm{der}}$, which is equivalent to x_n being regular in ℓ^{der}. □

Example 2.65. We may assume that $y \in \mathfrak{sl}_n$ is already in Jordan normal form. Then in fact it is regular if and only if all Jordan blocks have distinct eigenvalues. Here, the Levi subalgebra is given by block-diagonal matrices where the blocks are grouped by shared eigenvalues. However, any regular nilpotent element in $\mathfrak{sl}_m$ has a single Jordan block, so we see that each eigenvalue of y must correspond to a single Jordan block.

2.8. **Flag varieties and the Bruhat decomposition.** We will now discuss Borel subalgebras $\mathfrak{b} \subset \mathfrak{g}$.

Definition 2.66. A *Borel subalgebra* in $\mathfrak{g}$ is a maximal solvable subalgebra. A *Borel subgroup* in G is a maximal connected solvable subgroup.

Note that if we have a Borel subalgebra, we can integrate it to get a Borel subgroup $B \subset G$. Here are some properties:

(1) All Borel subalgebras are conjugate under the adjoint representation;
(2) If $\mathfrak{b}$ is a Borel subalgebra, then $\mathfrak{n}_{\mathfrak{g}}(\mathfrak{b}) = \mathfrak{b}$ and $N_G(\mathfrak{b}) = B$;
(3) Every Borel subalgebra contains a Cartan subalgebra. On the other hand, a fixed Cartan is contained in $|W_T|$-many Borel subalgebras, which form a W_T-torsor. In fact, they are in bijection with the set of choices of positive roots via
$$\Phi^+ \mapsto \mathfrak{b} = \mathfrak{h} \oplus \bigoplus_{\alpha \in \Phi^+} \mathfrak{g}_\alpha.$$
For example, we can let $\mathfrak{h}$ be the diagonal matrices in $\mathfrak{sl}_n$, and we can generate all Borels by taking the upper-triangular matrices as a base point and conjugating by permutation matrices.
(4) If $h \in \mathfrak{b}$ is regular semisimple, then in fact $\mathfrak{h} := Z_{\mathfrak{g}}(h) \subset \mathfrak{b}$.

Now let $\mathscr{B}$ be the **set** of Borel subalgebras of $\mathfrak{g}$. We can give it the structure of an algebraic variety as follows. First, consider the Grassmannian

$$\mathrm{Gr}(\dim \mathfrak{b} = r + \#\Phi^+, \mathfrak{g}),$$

which is naturally an algebraic variety which carries a G-action induced from the adjoint representation. Then $\mathscr{B}$ sits inside as a subset which is in fact the orbit of any Borel $\mathfrak{b}$ under this action. This gives

$$\begin{aligned}\mathscr{B} &= G \cdot \mathfrak{b} \\ &\cong G/N_G(\mathfrak{b}) \\ &\cong G/B.\end{aligned}$$

It is easy to check that $\mathscr{B}$ is in fact closed, so it is a projective variety called the *flag variety* of G (and the quotient description means it is a *homogeneous space*).

Remark 2.67. Because the Borel subgroup B always contains the center $Z(G)$, the flag variety is insensitive to the choice of intermediate form of G. If G is reductive, then this construction kills the center, so we only see the semisimple part of G in the flag variety.

Example 2.68. If $G = \mathrm{SL}_n$ (or PGL_n), then in fact

$$\mathscr{B} = \{0 \subsetneq V_1 \subsetneq V_2 \subsetneq \cdots \subsetneq V_n = \mathbb{C}^n \mid \dim V_i = i\}$$

is the usual flag variety of $\mathbb{C}^n$. This clearly admits a GL_n-equivariant surjection from the variety of frames

$$\{\{v_1, \dots, v_n\} \text{ is a basis of } \mathbb{C}^n\} \cong \mathrm{GL}_n\}.$$

If we consider the standard flag

$$\langle e_1 \rangle \subset \langle e_1, e_2 \rangle \subset \cdots \subset \langle e_1, \dots, e_n \rangle = \mathbb{C}^n,$$

then the stabilizer of this flag is the standard Borel subgroup B_+ of upper-triangular matrices, so we see that $\mathscr{B} = G/B_+$.

Now we may restrict the G-action on $\mathscr{B}$ to the Borel B, where we choose a particular Borel B as a base point. Then the set of B-orbits is denoted by

$$\{B\text{-orbits in } \mathscr{B}\} \cong B\backslash G/B.$$

We may also consider the $G \times G$-action on $\mathscr{B} \times \mathscr{B}$ and restrict to the diagonal subgroup G_Δ. Then the set of G_Δ-orbits is given by

$$B\backslash G/B \xrightarrow{\sim} \{G_\Delta\text{-orbits in } \mathscr{B} \times \mathscr{B}\} \qquad B \cdot \mathfrak{b}' \mapsto G_\Delta \cdot (\mathfrak{b}, \mathfrak{b}').$$

To check that this is well-defined, suppose that $\mathfrak{b}'' = b \cdot \mathfrak{b}'$. This follows from the fact that

$$\begin{aligned} G_\Delta \cdot (\mathfrak{b}, \mathfrak{b}'') &= G_\Delta \cdot (b \cdot (\mathfrak{b}, \mathfrak{b}')) \\ &= G_\Delta \cdot (\mathfrak{b}, \mathfrak{b}'). \end{aligned}$$

To check surjectivity, we look at the orbit of $(\mathfrak{b}_1, \mathfrak{b}_2)$, but we can bring $\mathfrak{b}_1$ to $\mathfrak{b}$, and surjectivity follows. To check injectivity, we see that if $G_\Delta \cdot (\mathfrak{b}, \mathfrak{b}') = G_\Delta(\mathfrak{b}, \mathfrak{b}'')$, then $(\mathfrak{b}, \mathfrak{b}'') = g \cdot (\mathfrak{b}, \mathfrak{b}')$ for some $g \in G$. But if $g \cdot \mathfrak{b} = \mathfrak{b}$, then $g \in B$ and thus $\mathfrak{b}'' \in B \cdot \mathfrak{b}'$.

We will now consider a more refined bijection which will give the Bruhat decomposition. Fix a maximal torus $T \subset B$, or equivalently a Cartan subalgebra $\mathfrak{h} \subset \mathfrak{b}$. Then we have a map

$$W_T \to B\backslash G/B, \qquad \dot{w} \mapsto B\dot{w} \cdot B,$$

where we choose a representative $\dot{w} \in wT$ of a coset in $W_T = N_G(T)/T$. Because $T \cdot B = B$, this map is well-defined. Taking stock, we now have maps

$$W_T \to B\backslash G/B \xrightarrow{\sim} \{B\text{-orbits in } \mathscr{B}\} \xrightarrow{\sim} \{G_\Delta\text{-orbits in } \mathscr{B} \times \mathscr{B}\}.$$

Our goal is to show that the final arrow is a bijection.

Theorem 2.69 (Bruhat decomposition)**.** *The map $W_T \to B\backslash G/B$ is a bijection.*

Remark 2.70. This implies the decomposition

$$G = \bigsqcup_{w \in W_T} B\dot{w}B.$$

Example 2.71. If $G = \mathrm{SL}_n$ and we choose B is the upper-triangular matrices with T the diagonal matrices, then we can present $\dot{w}$ as a permutation matrix, and the Bruhat decomposition says that any invertible matrix can be written as a product of an upper-triangular matrix, a permutation matrix, and another upper-triangular matrix.

The proof of this theorem is based on the Bialynicki-Birula decomposition, which is an analog of Morse theory. If we have an action of $\mathbb{C}^\times$ on a projective variety X, then we may consider the fixed point locus $\mathscr{W} = X^{\mathbb{C}^\times}$. We will assume for now that $\mathscr{W}$ is discrete (hence finite).

Example 2.72. If $X = \mathbb{P}^1$, then we consider the action $\lambda \cdot [a : b] = [a : \lambda b]$ with fixed points $0 = [1 : 0]$ and $\infty = [0 : 1]$. For any other point, we consider the limit

$$\lim_{\lambda \to 0} \lambda \cdot [a, b] = \lim_{\lambda \to 0} [a : \lambda b] = [1 : 0] = 0.$$

In this case, the attracting locus of 0 is $\mathbb{A}^1 = \mathbb{C}$, while the attracting locus of ∞ is just the point ∞.

For each fixed point $w \in \mathscr{W}$, we consider the tangent space $T_w X$, which is a representation of $\mathbb{C}^\times$. We can decompose it into weight spaces

$$T_w X = \bigoplus_{n \in \mathbb{Z}} T_w X[n],$$

where $T_w X[n]$ is the subspace on which $\lambda \in \mathbb{C}^\times$ acts by multiplication by λ^n. By assumption, there are no zero weights, so we have the decomposition

$$T_w X = T_w^+ X \oplus T_w^- X,$$

where the first summand is the sum of positive weight spaces and the second summand is the sum of negative weight spaces.

Theorem 2.73 (Bialynicki-Birula decomposition)**.** *The variety X decomposes into a disjoint union*

$$X = \bigsqcup_{w \in \mathscr{W}} X_w,$$

where the attracting manifold X_w is the locally closed subvariety

$$X_w \cong T_w X_w \cong T_w^+ X$$

and is also given by

$$X_w = \left\{ x \in X \,\middle|\, \lim_{\lambda \to 0} \lambda \cdot x = w \right\}.$$

Proof of Bruhat decomposition. We will establish bijections

$$\begin{aligned} W_T &\leftrightarrow \left\{ T\text{-fixed points in } \mathscr{B} \right\} \\ &\leftrightarrow \{ \mathbb{C}^\times\text{-orbits in } \mathscr{B} \text{ for some nice } \mathbb{C}^\times \hookrightarrow T \} \\ &\leftrightarrow \left\{ \text{attracting manifolds of the fixed points} \right\} \\ &\leftrightarrow \{ B\text{-orbits in } \mathscr{B} \}. \end{aligned}$$

Note that the third bijection follows directly from the Bialynicki-Birula decomposition, so we need to establish the other bijections.

First note that there is a bijection between T-fixed points in $\mathscr{B}$ and the set of all Borels $\mathfrak{b}' \supset \mathfrak{h}$. This is simply the infinitesimal version of the fact that if $T \cdot B' = B'$,

then $T \subset N_G(B') = B'$. Now this is in bijection with the set of $\dot{w} \cdot \mathfrak{b}$ for $w \in W_T$, which establishes the first bijection.

Now choose a cocharacter $h \colon \mathbb{C}^\times \hookrightarrow T$ which is regular and a primitive nonzero cocharacter of T. For this $\mathbb{C}^\times$-action, the fixed points are in bijection to the set of all Borel subgroups B' containing $\mathbb{C}^\times$. Taking the derivative at the identity, this is in bijection with

$$\{\mathfrak{b}' \mid \mathbb{C}h \subset \mathfrak{b}'\}.$$

Because h is regular, then $\mathfrak{h} = Z_{\mathfrak{g}}(h) \subset \mathfrak{b}'$, so we have recovered the set of all $\mathfrak{b}' \supset \mathfrak{h}$.

It remains to identify B-orbits with attracting sets of the $\mathbb{C}^\times$-fixed points. If $\mathfrak{b}_w \in \mathscr{B}^{\mathbb{C}^\times}$ is a Borel, then we consider its attracting set

$$\mathscr{B}_{\mathfrak{b}_w} = \left\{ \mathfrak{b}' \in \mathscr{B} \,\middle|\, \lim_{\lambda \to 0} \lambda \cdot \mathfrak{b}' = \mathfrak{b}_2 \right\}.$$

The attracting sets are the positive parts of the tangent spaces at the fixed points, and so we must have $T_{\mathfrak{b}_w}\mathscr{B} \cong \mathfrak{g}/\mathfrak{b}_w$ because $\mathscr{B} \cong G/B_w$. Note that the action of $\mathbb{C}^\times$ on $\mathfrak{g}$ decomposes into eigenspaces

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_n \mathfrak{g}_n,$$

where $\mathfrak{h}$ will disappear when we quotient by $\mathfrak{b}_w$. Our goal is to have

$$\mathfrak{g}_{>0} = \mathfrak{n} = [\mathfrak{b}, \mathfrak{b}].$$

Considering the root decomposition

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in \Phi} \mathfrak{g}_\alpha,$$

considering $\mathbb{C}^\times$-weights gives us

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha : \langle \alpha, h \rangle > 0} \mathfrak{g}_\alpha \oplus \bigoplus_{\alpha : \langle \alpha, h \rangle < 0} \mathfrak{g}_\alpha.$$

There is a Weyl chamber (see Figure 6 for the SL_3 example) which pairs with the positive roots Φ^+ positively, so we are free to choose $\mathfrak{h}$ in this chamber. Once we choose $\mathfrak{h}$, then the inclusion $\mathfrak{n}^+ \to \mathfrak{g}/\mathfrak{b}_w$ factors through $T_{\mathfrak{b}_w}\mathscr{B}$.

Remark 2.74. One convenient choice for a regular cocharacter h is $h = 2\rho^\vee$, where

$$\rho^\vee = \sum_{j \in \Pi} \omega_j^\vee = \frac{1}{2} \sum_{\beta \in \Phi^+} \beta^\vee.$$

This implies that

$$h = \sum_{\beta \in \Phi^+} \beta^\vee,$$

which lies in the coroot lattice $Q^\vee = \mathbb{X}_\bullet(T_{\mathrm{sc}})$. By definition, we have $\left\langle \rho^\vee, \alpha_j \right\rangle = 1$ for all simple roots, so it pairs positively with all positive roots.

Now let $U \subset [B, B]$ such that $\operatorname{Lie} U = \mathfrak{n}^+$. Then if we consider the orbit $U \cdot \mathfrak{b}_w$, we have

$$\mathfrak{n}^+ \to T_{\mathfrak{b}_w}(U \cdot \mathfrak{b}_w) \subset \mathfrak{g}/\mathfrak{b}_w.$$

Dimension counting shows that $U \cdot \mathfrak{b}_w$ is open and dense in $\mathscr{B}_{\mathfrak{b}_w}$, so our goal is to show that it is also closed, i.e. that $U \cdot \mathfrak{b}_w = \mathscr{B}_{\mathfrak{b}_w}$.

Lemma 2.75. *Suppose that U is a unipotent group acting on an affine variety $X = \operatorname{Spec} A$. Then any U-orbit in X is closed.*

Proof. Let $\mathscr{O}$ be a U-orbit and $\overline{\mathscr{O}}$ be its closure. Suppose that $C := \overline{\mathscr{O}} \smallsetminus \mathscr{O}$ is nonempty (and hence is a closed subvariety of X) and has defining ideal $0 \neq I \subset \mathbb{C}[\overline{\mathscr{O}}]$. Because C is U-stable, so is I.

Lemma 2.76. *For all $f \in I$, the span $\operatorname{Span}_{\mathbb{C}} U \cdot f$ is finite-dimensional.*

Proof. The action of U on X gives a coaction

$$A \to A \otimes \mathbb{C}[U] \qquad a \mapsto \sum_{i=1}^{m} a_i \otimes h_i.$$

Then we see that

$$u \cdot a = \sum_{i=1}^{m} a_i \otimes h_i(u)$$

and thus

$$\operatorname{Span}_{\mathbb{C}}\langle U \cdot a\rangle \subset \operatorname{Span}_{\mathbb{C}}\langle a_1, \dots, a_m\rangle,$$

which is finite-dimensional. □

Now consider $I^U \supset (\operatorname{Span}_{a\neq 0} U \cdot a)^U \neq 0$. Choose $0 \neq f \in I^U$. Then f is U-invariant, so it must be constant on $\mathscr{O}$, and hence constant on $\overline{\mathscr{O}}$. But we know that f vanishes on C, so it must vanish on $\overline{\mathscr{O}}$, which contradicts the fact that f is nonzero. Therefore, C must be empty and $\mathscr{O}$ is closed. □

The two lemmas together emply that $U \cdot \mathfrak{b}_w = B \cdot \mathfrak{b}_w = \mathscr{B}_{\mathfrak{b}_w}$. □

Example 2.77. In the case when $G = \mathrm{SL}_2$, then the flag variety if $\mathbb{P}^1$, and if we choose $\mathfrak{b}$ to be the upper-triangular matrices with $\mathfrak{h}$ the diagonal matrices, then the Weyl group is

$$W = \left\{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}\right\},$$

and the nontrivial element w swaps the upper-triangular and lower-triangular Borel subgroups. See Figure 8.

Note that the Bruhat decomposition even refines to

$$\mathscr{B} = \bigsqcup_{w \in W_T} U\dot{w} \cdot B$$

and gives a cell decomposition of $\mathscr{B}$ into affine spaces (which are even dimensional). This allows us to easily compute the homology and cohomology of $\mathscr{B}$, which is given by

$$H_\bullet(\mathscr{B}) = \bigoplus_{w \in W} ([U\dot{w} \cdot B])[2 \dim U\dot{w} \cdot B].$$

We can compute the dimension of each cell as follows. Note that

$$U\dot{w} \cdot B = (U\dot{w} \cap \dot{w}U^-) \cdot B,$$

where U^- is the unipotent radical of the opposite Borel. Now we compute

$$\begin{aligned} \dim \mathscr{B}_w &= \dim(U\dot{w} \cap wU^-) \\ &= \dim(\dot{w}^{-1}U\dot{w} \cap U^-) \end{aligned}$$

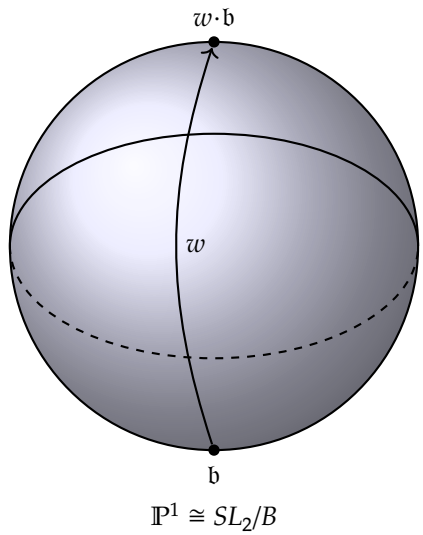


FIGURE 8. Action of Weyl group on Borels for SL_2.

$$= \dim(\mathrm{Ad}_{\dot{w}^{-1}}\mathfrak{n} \cap \mathfrak{n}^-)$$
$$= \$w^{-1}(\Phi^+) \cap (-\Phi^+) =: \ell(w),$$

which is called the *length* of w relative to Φ^+. For example, the identity has length 0 and the longest element w_0 has length $\#\Phi^+$.

Example 2.78. In the SL_n case, the flag variety has dimension

$$|\Phi^+| = 1 + \cdots + (n-1) = \frac{n(n-1)}{2}.$$

The cohomology has dimension

$$\dim H^\bullet(\mathscr{B}) = |W| = n!.$$

Later, we will see that this is actually a representation of the Weyl group W and describe its algebra structure.

Remark 2.79. In fact, a Borel subgroup $B \subset G$ is a minimal subgroup of G such that the quotient G/B is projective.

Theorem 2.80 (Fixed point theorem of Borel)**.** *A connected solvable linear algebraic group acting algebraically on a non-empty complete variety has a fixed point.*

2.9. **Parabolic subgroups.** If $X = G/H$ is a homogeneous space which is compact, we can consider the action of a Borel subgroup $B \subset G$ on X. By Borel's fixed point theorem, this has a fixed point, say gH. But then the stabilizer gHg^{-1} of this point contains B, and therefore

$$g^{-1}Bg \subset H$$

so H contains a Borel. Conversely, if H contains a Borel, then G/H admits a surjection from the flag variety G/B and is hence projective.

Definition 2.81. A *parabolic subgroup* of G is a (necessarily connected) subgroup $P \subset G$ which contains a Borel subgroup.

Remark 2.82. We can drop connectedness from the definition because P is in fact always connected.

Proposition 2.83. *A subgroup $P \subset G$ is parabolic if and only if G/P is projective.*

We can in fact "classify" parabolic subgroups. If we fix a Borel B, then we can ask for all parabolic subgroups containing B. Because P is connected, we can reduce to the corresponding problem for Lie algebras, namely finding $\mathfrak{p} \supset \mathfrak{b}$ such that $[\mathfrak{p}, \mathfrak{p}] \subset \mathfrak{p}$. We may consider the adjoint action of $\mathfrak{h}$ on $\mathfrak{p}$, which decomposes as

$$\mathfrak{p} = \mathfrak{h} \oplus \bigoplus_{\alpha \in \Phi^+} \mathfrak{g}_\alpha \oplus \bigoplus_{\beta \in \Gamma} \mathfrak{g}_\beta.$$

The first two factors give $\mathfrak{b}$, so our goal is to determine the possible Γ. We will not discuss the details and only state the end result.

Proposition 2.84. *There is a one-to-one correspondence between $\mathfrak{p} \supset \mathfrak{b}$ and subsets $S \subset \Pi$ of the simple roots. It is given by*

$$\mathfrak{p} \supset \mathfrak{b} \mapsto S = S = -\Gamma \cap \Pi.$$

If we restrict the Cartan matrix to S, we obtain a root system

$$S = S_1 \sqcup S_2 \sqcup \cdots \sqcup S_k,$$

which decomposes into irreducible root systems. We can then decompose

$$\mathfrak{p} = \mathfrak{p}^u \oplus \ell,$$

where $\mathfrak{p}^u$ is the nilradical and ℓ is the Levi factor, then $\ell^{\mathrm{ss}} = \ell/Z(\ell)$ has root system determined by S. The center is given by

$$\begin{aligned} Z(\ell) &= S^\perp \\ &= \{h \in \mathfrak{h} \mid \alpha(h) = 0 \text{ for all } \alpha \in S\} \\ &= \bigcap_{\alpha \in S} \ker(\alpha). \end{aligned}$$

In particular, we have

$$\ell = \mathfrak{h} \oplus \bigoplus_{\alpha \in \Phi_S} \mathfrak{g}_\alpha.$$

For example, in the SL_3 example, if we choose S to be one simple root, then we see only roots living in the span of that simple root, which gives us a copy of $\mathfrak{sl}_2$.

Example 2.85. If $\mathfrak{g} = \mathfrak{sl}_n$ and $\mathfrak{b} = \mathfrak{b}_n$ is the upper-triangular matrices, then the parabolic subalgebras are simply block upper-triangular matrices and the Levi factor is given by block-diagonal matrices. The root system of the Levi is determined by the sizes of the blocks, and the center is given by scalar matrices in each block.

Remark 2.86. The varieties G/P are known as *partial flag varieties*. In the $G = \mathrm{SL}_n$ example, they are given by flags

$$0 = V_0 \subset V_1 \subset V_2 \subset \cdots \subset V_k = \mathbb{C}^n,$$

where we require that $\dim V_i/V_{i-1} = n_i$ is the size of the i-th block.

2.10. **Representation theory.** We will give a classification of irreducible representations of $\mathfrak{g}$.

Theorem 2.87. *Fixing $\mathfrak{h} \subset \mathfrak{b} \subset \mathfrak{g}$, the (isomorphism classes of) irreducible representations of $\mathfrak{g}$ are in one-to-one correspondence with dominant weights $\lambda \in \Lambda$, where*

$$\begin{aligned}\Lambda &= \mathrm{Span}_{i\in\Pi}\{\omega_i\} \\ &= \left\{\lambda \in \mathfrak{h}^* \mid \langle \lambda, \alpha^\vee \rangle \in \mathbb{Z}_{\geq 0} \text{ for all } \alpha \in \Phi^+\right\}\end{aligned}$$

is the nonnegative integral span of the fundamental weights.

In the $\mathfrak{sl}_3$ example, this corresponds to the shaded region in Figure 5.

Remark 2.88. The case of groups is more subtle because of fundamental group issues. However, representations of $\mathfrak{g}$ are "the same" as representations of the simply connected form G_{sc}. This happens because we can integrate $\mathfrak{g} \to \mathfrak{gl}(V)$ to $G_{\mathrm{sc}} \to \mathrm{GL}(V)$. In particular, this means that irreducible representations of G_{sc} are the same as positive coweights $\mathbb{X}^\bullet(T_{\mathrm{sc}})^+ = \Lambda^+$.

The way we do this is by considering the restriction to $\mathfrak{h}$ for any $\mathfrak{g}$-representation. These are all semisimple by preservation of Jordan decomposition. This gives a decomposition

$$V = \bigoplus_{\mu\in\mathfrak{h}^*} V_\mu,$$

so because we can integrate to G_{sc}, the $\mathfrak{h}$-action upgrades to a T_{sc}-action, and we have a decomposition

$$V = \bigoplus_{\mu\in\mathbb{X}^\bullet(T_{\mathrm{sc}})} V_\mu.$$

We can now consider the action of the unipotent radical $U \subset B$ on V, which will not preserve the weight decomposition.

U has a nonzero fixed point because the B-action on $\mathbb{P}(V)$ has a fixed point by Borel's fixed point theorem. If we consider the action of U on V_μ, then

$$U \cdot v_\mu = v_\mu + \sum_{q\in Q^+} v'_{\mu+q},$$

where Q^+ is the positive part of the root lattice. We will say that $\mu \leq \nu$ if $\nu - \mu \in Q^+$. In this language, we have

$$U \cdot v_\mu = v_\mu + \sum_{\nu \gtrsim \mu} v'_\nu.$$

Definition 2.89. A *highest weight vector* v_λ of V is a nonzero vector such that $U \cdot v_\lambda = v_\lambda$.

Theorem 2.90. *Every irreducible representation V has a unique highest weight vector v_λ up to scaling, and this λ is called the highest weight. Conversely, V is uniquely determined by its highest weight.*

For an arbitrary finite-dimensional representation V, we can compute the decomposition

$$V = \bigoplus_{i,j} V_{\lambda_i}^{(j)}$$

by considering the action of U on V. To decompose, we look for the fixed subspace V^U, which is the span of the highest weight vectors. Finite-dimensional representations of B are classified by $\mathbb{X}^\bullet(B_{\mathrm{sc}}/B^u_{\mathrm{sc}}) = \mathbb{X}^\bullet(T_{\mathrm{sc}}) = \Lambda$. For all $\lambda \in \Lambda$, let $\mathbb{C}_\lambda$ be the corresponding irreducible representation of B. Then the associated line bundle

$$\begin{aligned}\mathscr{L}_\lambda &= G \times^B \mathbb{C}_{-\lambda} \\ &= (G \times \mathbb{C}_{-\lambda})/B \\ &= \left\{(g, v) \sim (gb^{-1}, b \cdot v)\right\}\end{aligned}$$

on G/B is G-equivariant.

Theorem 2.91 (Borel-Weil)**.** *The global sections $\Gamma(\mathscr{B} = G/B, \mathscr{L}_\lambda)$ are isomorphic to V^*_λ as G_{sc}-representations.*

Remark 2.92. There is a generalization of this result, called the Borel-Weil-Bott theorem, which calculates the higher cohomology of these line bundles.

We will not give a detailed proof of the Borel-Weil theorem. However, the idea is that a section of $\mathscr{L}_\lambda$ is a pair $(g, f(g) \cdot 1)$ where $f \colon G_{\mathrm{sc}} \to \mathbb{C}$ (here, we always use the simply connected form and drop the subscripts), and we must be invariant under the B-action

$$(g, f(g) \cdot 1) \sim (gb^{-1}, \lambda(b)^{-1} f(g) \cdot 1) = (gb^{-1}, f(gb^{-1}) \cdot 1),$$

and therefore we must have

$$f(gb^{-1}) = \lambda(b)^{-1} f(g)$$

for all $g \in G$ and $b \in B$. Equivalently, we can ask for

$$f(gb) = \lambda(b) f(g).$$

Then we are looking for $f \in \mathscr{O}(G)^{(B,\lambda)} \cong \Gamma(\mathscr{B}, \mathscr{L}_\lambda)$. To identify this representation, we need to find the highest weight with respect to B or B^-. It is a bit easier to work with B^-, so we can pass to U^-. Then we consider

$$\mathscr{O}(G)^{U^- \times (B,\lambda)} \cong \mathscr{O}(G)^{U^- \times U, (T,\lambda)}.$$

Using the Bruhat decomposition

$$G = \bigsqcup_{w \in W} B\dot{w}B = \bigsqcup_{w \in W} B^-\dot{w}B,$$

we will use the longest element w_0 (for SL_n this is the permutation which exchanges 1 and n, 2 and $n-1$, etc.) to write

$$G = \bigsqcup_{w \in W} \dot{w}_0 B \dot{w}_0 \dot{w} B.$$

Then $B^-B \subset G$ is open, so we can consider the behavior of functions on this open. This decomposes as the Cartesian product

$$B^-B = C^-TU \cong U^- \times T \times U,$$

so we are looking for functions in $\mathscr{O}(U^-TU)^{U^- \times U, (T,\lambda)} = \mathscr{O}(T)^{(T,\lambda)}$. This gives us the 1-dimensional space $\mathbb{C} \cdot y^\lambda$, so the highest weight space we are trying to calculate is at most 1-dimensional. It remains to show that y^λ has no poles outside of B^-B. Using the Bruhat decomposition again, the codimension 1 orbits with respect to $B^- \times B$ are given by $B^-\dot{s}_i B$ for simple reflections s_i.

Lemma 2.93. *For all simple coroots $\alpha_i^\vee$, let ω_i be the dual fundamental weight. Then* $\operatorname{div}(y^{\omega_i}) = \overline{B^- \dot{s}_i B}$.

Now if $\lambda \in \Lambda^+$, we see that y^λ is globally defined and that

$$\mathscr{O}(G)^{U^- \times U, (T,\lambda)} = \mathbb{C} \cdot y^\lambda.$$

If $\lambda \notin \Lambda^+$, then

$$\lambda = \sum_{i=1}^r a_i \omega_i$$

with some of the a_i being negative, so y^λ has a pole along D_i of order $|a_i|$. Thus, B^- acts on $\mathscr{O}(G)^{(B,\lambda)}$ with weight $-\lambda$, and the end result is that the representation must be V_λ^*.

Remark 2.94. Note that $\operatorname{Pic}(G) = 0$ because of the Bruhat decomposition

$$G = U^- T U \cup \bigcup_{i=1}^r \overline{B^- \dot{s}_i B},$$

and therefore the the exact sequence

$$\bigoplus_{i=1}^r \mathbb{Z}[D_i] \to \operatorname{Pic}(G) \to \operatorname{Pic}(U^- T U) = 0$$

has the first object given entirely by principal divisors $D_i = \overline{B^- \dot{s}_i B}$. Equivalently, we see that $\mathscr{O}(G)$ is a UFD.

However, if we move away from the simply-connected form, then there are nontrivial line bundles, and

$$\operatorname{Pic}(G) \cong \pi_1(G).$$

Theorem 2.95 (Peter-Weyl)**.** *Consider the action of $G_{\mathrm{sc}} \times G_{\mathrm{sc}}$ on $\mathscr{O}(G_{\mathrm{sc}})$. Then we have a decomposition*

$$\mathscr{O}(G_{\mathrm{sc}}) = \bigoplus_{\lambda \in \Lambda^+} V_\lambda^* \otimes V_\lambda.$$

We may obtain this decomposition by considering $U^- \times U$-fixed vectors and obtaining

$$\mathscr{O}(G_{\mathrm{sc}})^{U^- \times U} = \bigoplus_{\lambda \in \Lambda^+} \mathbb{C} y^\lambda,$$

which has weight $-\lambda$ for B^- in the first factor and weight λ for B in the second factor. In the other direction, we associate the tensor $v_1^* \otimes v_2$ with the function

$$f_{v_1^* \otimes v_2}(g) = \langle v_1^*, g \cdot v_2 \rangle.$$

3. Springer theory

3.1. The Grothendieck-Springer resolution.

Proposition 3.1. *Let $\mathfrak{b}$ and $\mathfrak{b}'$ be two Borel subalgebras of $\mathfrak{g}$. Then there is a canonical isomorphism $\mathfrak{b}/[\mathfrak{b},\mathfrak{b}] \xrightarrow{\sim} \mathfrak{b}'/[\mathfrak{b}',\mathfrak{b}']$.*

Proof. Because all Borels are conjugate, we have

$$\mathfrak{b}' = \mathrm{Ad}_g\, \mathfrak{b}.$$

This isomorphism preserves the nilradical, so it induces an isomorphism on the quotients. For a different g' that conjugates $\mathfrak{b}$ to $\mathfrak{b}'$, we have $g' = gb$ for some $b \in B$, and it remains to check that

$$\mathfrak{b}/[\mathfrak{b},\mathfrak{b}] \xrightarrow{\mathrm{Ad}_b} \mathfrak{b}/[\mathfrak{b},\mathfrak{b}] \xrightarrow{\mathrm{Ad}_g} \mathfrak{b}'/[\mathfrak{b}',\mathfrak{b}']$$

is the same as

$$\mathfrak{b}/[\mathfrak{b},\mathfrak{b}] \xrightarrow{\mathrm{Ad}_g} \mathfrak{b}'/[\mathfrak{b}',\mathfrak{b}'].$$

Passing to the infinitesimal version, the action of $\mathfrak{b}$ on $\mathfrak{b}/[\mathfrak{b},\mathfrak{b}]$ is zero because the adjoint action of $\mathfrak{b}$ on itself sends everything to the nilradical, so it integrates to the trivial action of B. □

Definition 3.2. The quotient $\mathfrak{b}/[\mathfrak{b},\mathfrak{b}]$ is called the *universal Cartan* of $\mathfrak{g}$ and is denoted by $\mathfrak{H}$.

Remark 3.3. We must note that $\mathfrak{H}$ is **not** a subalgebra of $\mathfrak{g}$. Instead, we should view it as a pair $(\mathfrak{h}, \mathfrak{b} \subset \mathfrak{h})$. Indeed, for all such pairs, we can identify $\mathfrak{h}$ and $\mathfrak{H}$ by

$$\mathfrak{h} \hookrightarrow \mathfrak{b} \twoheadrightarrow \mathfrak{b}/[\mathfrak{b},\mathfrak{b}] = \mathfrak{H}.$$

If $\mathfrak{h} \subset \mathfrak{b}'$ is a different Borel, then $\mathfrak{b}' = w(\mathfrak{b}) = \dot{w} \cdot \mathfrak{b}$, and therefore relating the sequences via

$$\begin{tikzcd}
\mathfrak{h} \arrow[r, hook] \arrow[d, "\mathrm{Ad}_{\dot{w}}"] & \mathfrak{b} \arrow[r, two heads] \arrow[d, "\mathrm{Ad}_{\dot{w}}"] & \mathfrak{b}/[\mathfrak{b},\mathfrak{b}] \arrow[r, equal] \arrow[d, "\cong"] & \mathfrak{H} \arrow[d, equal] \\
\mathfrak{h} \arrow[r, hook] & \mathfrak{b}' \arrow[r, two heads] & \mathfrak{b}'/[\mathfrak{b}',\mathfrak{b}'] \arrow[r, equal] & \mathfrak{H}
\end{tikzcd}$$

requires conjugation by $\dot{w}$ and thus requires the choice of a Borel. However, we do end up with a pair $(\mathfrak{H}, \Phi^+_{\mathfrak{H}})$, which gives a universal set of positive roots containing a universal set $\Pi_{\mathfrak{H}}$ of simple roots. We will call this a "pinned" version of the usual Cartan. This allows us to discuss the Weyl group $\mathbb{W}$ and the universal simple reflections. Here, we note that any choice of $\mathfrak{h} \subset \mathfrak{b}$ gives a canonical isomorphism

$$W_T \xrightarrow{\sim} \mathbb{W}.$$

Lemma 3.4. *For any Borel subalgebras $\mathfrak{b}_1$ and $\mathfrak{b}_2$ there exists a Cartan subalgebra $\mathfrak{h} \subset \mathfrak{b}_1 \cap \mathfrak{b}_2$.*

Proof. Using the Bruhat decomposition relative to $B_1 \supset T_1$, we write

$$\mathscr{B} = \bigsqcup_{w \in W} B_1 \dot{w} \cdot \mathfrak{b}_1.$$

There exists some $u_1 \in U_1$ and $w \in W_{T_1}$ such that

$$\mathfrak{b}_2 = u_1 \dot{w} \cdot \mathfrak{b}_1.$$

We consider $u_1^{-1} \cdot \mathfrak{b}_2 = \dot{w} \cdot \mathfrak{b}_1$ and $u_1^{-1} \cdot \mathfrak{b}_1 = \mathfrak{b}_1$. In particular, we see that

$$\mathfrak{h}_1 \subset u_1^{-1} \cdot \mathfrak{b}_2 \cap u_1^{-1} \cdot \mathfrak{h}_1.$$

Applying u_1 to this inclusion gives us the desired result. □

Remark 3.5. If we consider the set of pairs $\{(\mathfrak{h}, \mathfrak{b})\}$, this is actually isomorphic to G/T. Forgetting $\mathfrak{h}$ corresponds to the map $G/T \to G/B$, and the fiber over the set

$$\{\mathfrak{h} \subset \mathfrak{b}\} \cong B/T \cong U.$$

Definition 3.6. Define the *relative position* of two Borel subalgebras $\mathfrak{b}_1$ and $\mathfrak{b}_2$ as follows. Choose a Cartan $\mathfrak{h} \subset \mathfrak{b}_1 \cap \mathfrak{b}_2$. The pair $(\mathfrak{h}, \mathfrak{b}_1)$ determines a choice of positive roots, and we also know that $\mathfrak{b}_2 = \dot{w} \cdot \mathfrak{b}_1$. We then have canonical identifications $(\mathfrak{h}, \mathfrak{b}_1) \cong (\mathfrak{H}, \Phi_{\mathfrak{H}}^+)$ and $W_T \simeq \mathbb{W}$. Therefore, we have

$$\mathfrak{b}_1 \xrightarrow{w} \mathfrak{b}_2$$

for a unique $w \in \mathbb{W}$.

The reason this is well defined is because for any two $\mathfrak{h}_1, \mathfrak{h}_2 \subset \mathfrak{b}_1 \cap \mathfrak{b}_2$, we have $\mathfrak{h}_2 = b \cdot \mathfrak{h}_1$ for some $b \in B_1 \cap B_2$. This implies that in the diagram

$$\begin{tikzcd}
(\mathfrak{h}_1, \mathfrak{b}_1) \arrow[rr, "\mathrm{Ad}_b"] \arrow[dr] & & (\mathfrak{h}_2, \mathfrak{b}_1) \arrow[dl] \\
& (\mathfrak{H}, \Phi_{\mathfrak{H}}^+), &
\end{tikzcd}$$

we actually have $\dot{w}_2 = \mathrm{Ad}_b \, \dot{w}_1$, so they are sent to the same $w \in \mathbb{W}$.

In practice, if we fix a pair $(\mathfrak{h}, \mathfrak{b})$, then given $\mathfrak{b}_1$ and $\mathfrak{b}_2$, we first write $\mathfrak{b}_1 = u\dot{w} \cdot \mathfrak{b}$, then apply

$$(\mathfrak{b}_1, \mathfrak{b}_2) \xrightarrow{\mathrm{Ad}_{(u\dot{w})^{-1}}} (\mathfrak{b}, (u\dot{w})^{-1} \cdot \mathfrak{b}_2),$$

and then compute which Bruhat cell $(u\dot{w})^{-1} \cdot \mathfrak{b}$ lies in.

Remark 3.7.

(1) If we consider the action of G_Δ on $\mathscr{B} \times \mathscr{B}$, then the orbits are canonically indexed by $\mathbb{W}$.
(2) If we have $\mathfrak{b}_1 \xrightarrow{w_1} \mathfrak{b}_2 \xrightarrow{w_2} \mathfrak{b}_3$, the relative position of $\mathfrak{b}_3$ with respect to $\mathfrak{b}_1$ is **not** determined by w_1 and w_2. However, if $\ell(w_1) + \ell(w_2) = \ell(w_1 w_2)$, then the relative position is indeed $w_1 w_2$.

Example 3.8. If $G = \mathrm{SL}_2$, then $\mathscr{B} \times \mathscr{B} = \mathbb{P}^1 \times \mathbb{P}^1$. The G_Δ-action decomposes this into

$$\mathbb{P}^1 \times \mathbb{P}^1 = \Delta\mathbb{P}^1 \sqcup (\mathbb{P}^1 \times \mathbb{P}^1 \smallsetminus \Delta\mathbb{P}^1),$$

and the diagonal corresponds to the identity element (so $\mathfrak{b}_1 = \mathfrak{b}_2$), while the other element corresponds to the large open set. If we consider

$$\mathfrak{b}_1 \xrightarrow{s} \mathfrak{b}_2 \xrightarrow{s} \mathfrak{b}_3,$$

then we can get either 1 or s depending on whether $\mathfrak{b}_1 = \mathfrak{b}_3$.

In general, suppose $\mathfrak{b}_1, \mathfrak{b}_2, \mathfrak{b}_3 \supset \mathfrak{h}$ and

$$\mathfrak{b}_1 \xrightarrow{w_1} \mathfrak{b}_2 \xrightarrow{w_2} \mathfrak{b}_3.$$

Suppose that $\mathfrak{b}_1 = \mathfrak{b}$. Applying $\dot{w}^{-1}$ to the second arrow, we see that

$$\dot{w}_1^{-1} \xrightarrow{w_2} \dot{w}_1^{-1}\mathfrak{b}_3,$$

which tells us that $\dot{w}_1^{-1}\mathfrak{b}_3 = \dot{w}_2\mathfrak{b}$ and hence that $\mathfrak{b}_3 = \dot{w}_1\dot{w}_2\mathfrak{b}$.

Definition 3.9. The *Grothendieck-Springer resolution* is the incidence variety

$$\tilde{\mathfrak{g}} := \{(x, \mathfrak{b}) \in \mathfrak{g} \times \mathscr{B} \mid x \in \mathfrak{b}\}.$$

This variety admits two maps

$$\begin{tikzcd} & \tilde{\mathfrak{g}} \arrow[dl, "\mu"'] \arrow[dr, "\pi"] & \\ \mathfrak{g} & & \mathscr{B} \end{tikzcd}$$

which are both G-equivariant. Over $\mathscr{B}$, the fibers are simply given by $\mathfrak{b}$, so we have the tautological vector bundle over $\mathfrak{b}$. Therefore, we have a morphism

$$G \times^B \mathfrak{b} \to \tilde{\mathfrak{g}} \qquad (g, x) \mapsto (g \cdot x, g \cdot \mathfrak{b})$$

over $G/B \cong \mathscr{B}$.

The fiber

$$\mu^{-1}(x) = \{\mathfrak{b} \in \mathscr{B} \mid x \in \mathfrak{b}\}$$

which is the zero locus of the vector field on $\mathscr{B}$ corresponding to x is called the *Springer fiber*. Later, we will restrict to x in the nilpotent cone. The first extreme case is if $x = 0$, in which case $\mu^{-1}(0) = \mathscr{B}$. On the other hand, if $x \in \mathfrak{g}_{\mathrm{ss}}^{\mathrm{reg}}$, then $x \in \mathfrak{h}^{\mathrm{reg}}$ for some Cartan, so $\mathfrak{h} \subset \mathfrak{b}$ for any $\mathfrak{b} \ni x$, and thus the Springer fiber is simply

$$\mu^{-1}(x) = \{\mathfrak{b} \supset \mathfrak{h}\},$$

which is a torsor for the Weyl group. In particular, $\tilde{\mathfrak{g}}^{\mathrm{rs}}$ is actually a $\mathbb{W}$-torsor over $\mathfrak{g}^{\mathrm{rs}}$.

Proposition 3.10. *The map μ is proper.*

Proof. Consider the commutative diagram

$$\begin{tikzcd} \tilde{\mathfrak{g}} \arrow[r, hook] \arrow[dr, "\mu"'] & \mathfrak{g} \times \mathscr{B} \\ & \mathfrak{g} \end{tikzcd}$$

and use the fact that the horizontal arrow is a closed immersion and the fact that proper morphisms are closed under composition. □

Now consider the morphism

$$\nu \colon \tilde{\mathfrak{g}} \to \mathfrak{H} \qquad (x, \mathfrak{b}) \mapsto (x \mod [\mathfrak{b}, \mathfrak{b}]).$$

Note that if we consider $\mathfrak{g}$ only, then the target must be $\mathfrak{h} /\!/ W$ because there is no preferred order of the eigenvalues (in the $\mathfrak{sl}_n$ case, the others are similar). We now have the correspondence

$$\begin{tikzcd} & \tilde{\mathfrak{g}} \arrow[dl, "\mu"'] \arrow[dr, "\nu"] & \\ \mathfrak{g} & & \mathfrak{H}. \end{tikzcd} \tag{1}$$

Theorem 3.11 (Chevalley restriction theorem)**.** *Fix a Cartan $\mathfrak{h} \subset \mathfrak{g}$. Then there is the restriction of functions $\mathscr{O}(\mathfrak{g}) \to \mathscr{O}(\mathfrak{h})$, which induces*

$$\varphi \colon \mathscr{O}(\mathfrak{g})^G \to \mathscr{O}(\mathfrak{h})^W.$$

This φ is an isomorphism.

Proof. We begin with injectivity. If $P \in \mathscr{O}(\mathfrak{g})^G$ satisfies $P|_{\mathfrak{h}} = 0$, then $P|_{\mathfrak{g}^s} = 0$. This is dense, so $P = 0$.

Now let $P \in \mathscr{O}(\mathfrak{h})^W$. Our goal is to find $\tilde{P} \in \mathscr{O}(\mathfrak{g})^G$ such that $\tilde{P}|_{\mathfrak{h}} = P$. Note that P determines $\tilde{P}$ on $\mathfrak{g}^{\mathrm{rs}}$, but we need to ensure that $\tilde{P}$ extends to all of $\mathfrak{g}$ (since the boundary has codimension 1).

First, using the correspondence Equation (1), we have $P_{\mathfrak{H}} \in \mathscr{O}(\mathfrak{H})$ after we choose some $\mathfrak{b} \supset \mathfrak{h}$. Because P is W-invariant, this in fact does not depend on the choice of $\mathfrak{b}$. We now have a function $\tilde{P} = \nu^* P_{\mathfrak{H}}$ on $\tilde{\mathfrak{g}}$, which is G-invariant because

$$\begin{aligned}\tilde{P}(g \cdot x, g \cdot \mathfrak{b}) &= P_{\mathfrak{H}}(g \cdot x + [g \cdot \mathfrak{b}, g \cdot \mathfrak{b}]) \\ &= P_{\mathfrak{H}}(x + [\mathfrak{b}, \mathfrak{b}]) \\ &= \tilde{P}(x, \mathfrak{b}).\end{aligned}$$

Our goal now is to descend it to $\mathfrak{g}$. Let $x \in \mathfrak{g}^{\mathrm{rs}}$. Note that $\tilde{\mathfrak{g}}^{\mathrm{rs}} \to \mathfrak{g}^{\mathrm{rs}}$ is a $\mathbb{W}$-torsor and thus $\mathfrak{g}^{\mathrm{rs}} = \tilde{\mathfrak{g}}^{\mathrm{rs}}/\mathbb{W}$. In particular, we see that

$$\mathscr{O}(\mathfrak{g}^{\mathrm{rs}}) = \mathscr{O}(\tilde{\mathfrak{g}}^{\mathrm{rs}})^{\mathbb{W}}.$$

Note that $\tilde{P}|_{\tilde{\mathfrak{g}}}^{\mathrm{rs}}$ is $\mathbb{W}$-invariant, so it descends to $R \in \mathscr{O}(\mathfrak{g}^{\mathrm{rs}})^G$. Moving to the analytic topology, we want to show that for all $x \in \mathfrak{g} \smallsetminus \mathfrak{g}^{\mathrm{rs}}$ and any small ball $D \ni x$, then $R|_{D \cap \mathfrak{g}^{\mathrm{rs}}}$ is bounded. Note that $\mu^{-1}(\bar{D})$ is compact because μ is proper. Note that $\tilde{P}|_{\mu^{-1}(\bar{D}) \cap \tilde{\mathfrak{g}}^{\mathrm{rs}}}$ is bounded, so $R|_{\bar{D} \cap \mathfrak{g}^{\mathrm{rs}}}$ is bounded and thus extends to a holomorphic function on D.

Finally, we need to check that $R|_{\mathfrak{h}} = P$. For any $x \in \mathfrak{h}$ and any choice of $\mathfrak{b} \ni x$, we see that

$$\begin{aligned}R(x) &= \tilde{P}(x, \mathfrak{b}) \\ &= P_{\mathfrak{H}}(x + [\mathfrak{b}, \mathfrak{b}]) \\ &= P(x).\end{aligned}$$

□

Example 3.12. For $\mathfrak{g} = \mathfrak{sl}_n$, we can write

$$P_t(A) = t^n + 0 \cdot t^{n-1} + a_2(A)t^{n-2} + \cdots + a_n(A),$$

and $\mathscr{O}(\mathfrak{g})^G = \mathbb{C}[a_2, \dots, a_n]$. Transported to $\mathfrak{h}$, these are simply the elementary symmetric polynomials in the eigenvalues.

Note also that the Chevalley restriction theorem gives a canonical isomorphism $\mathscr{O}(\mathfrak{g})^G \simeq \mathscr{O}(\mathfrak{H})^{\mathbb{W}}$. Later, we will see that $\mathscr{O}(\mathfrak{H})^{\mathbb{W}}$ is a polynomial ring in r variables, so $\mathfrak{H} /\!/ \mathbb{W}$ is an affine space of dimension r. The embedding $\mathscr{O}(\mathfrak{H})^{\mathbb{W}} \hookrightarrow \mathscr{O}(\mathfrak{g})$ gives the adjoint quotient map

$$\chi \colon \mathfrak{g} \to \mathfrak{H} /\!/ \mathbb{W}.$$

Remark 3.13. We can in fact choose homogeneous generators of degrees $d_1 \le d_2 \le \cdots \le d_r$, and we will have $d_1 = 2$ and d_r being the Coxeter number h.

Proposition 3.14. *The diagram*

$$\begin{tikzcd} & \tilde{\mathfrak{g}} \arrow[dl, "\mu"'] \arrow[dr, "\nu"] & \\ \mathfrak{g} \arrow[dr, "\chi"] & & \mathfrak{H} \arrow[dl, "\pi"] \\ & \mathfrak{H} /\!/ \mathbb{W} & \end{tikzcd}$$

commutes.

Proof. It suffices to check that

$$\mu^* \chi^* \colon \mathscr{O}(\mathfrak{H} /\!/ \mathbb{W}) \to \mathscr{O}(\tilde{\mathfrak{g}}) = \nu^* \pi^*$$

because $\mathfrak{H} /\!/ \mathbb{W}$ is affine. This is true because

$$\begin{aligned} \mu^* \chi^*(P)(x, \mathfrak{b}) &= \nu^* P(x, \mathfrak{b}) \\ &= P(x + [\mathfrak{b}, \mathfrak{b}]) \end{aligned}$$

essentially by definition. □

Proposition 3.15. *Let $\mathfrak{b} \in \mathscr{B}$ be a Borel, $\mathfrak{n} = [\mathfrak{b}, \mathfrak{b}]$, and $x \in \mathfrak{b}$. Then for all $P \in \mathscr{O}(\mathfrak{g})^G$, the restriction $P|_{x+\mathfrak{n}}$ is constant.*

Proof. For all $n \in \mathfrak{n}$, we have

$$\begin{aligned} P(x + n) &= \tilde{P}(x + n, \mathfrak{b}) \\ &= P_{\mathfrak{H}}(x + n + [\mathfrak{b}, \mathfrak{b}]) \\ &= P_{\mathfrak{H}}(x + [\mathfrak{b}, \mathfrak{b}]) \\ &= \tilde{P}(x, \mathfrak{b}) \\ &= P(x). \end{aligned}$$

□

Remark 3.16. If we use the Jordan decomposition, then in fact $P(x) = P(x_s)$ for any $P \in \mathscr{O}(\mathfrak{g})^G$ because $P(x) = P_{\mathfrak{H}}(x_s)$.

3.2. **Nilpotent cone.**

Definition 3.17. The set of nilpotent elements

$$\mathscr{N} = \left\{ x \in \mathfrak{g} \mid x \text{ is nilpotent} \right\}$$

is closed, G-stable, and $\mathbb{C}^\times$-stable and is called the *nilpotent cone*.

Recalling the Grothendieck-Springer resolution $\mu \colon \tilde{\mathfrak{g}} \to \mathfrak{g}$, we define $\tilde{\mathscr{N}} = \mu^{-1}(\mathscr{N})$. Now we can consider the diagram

$$\begin{tikzcd} & \tilde{\mathscr{N}} \arrow[dl, "\mu_{\mathscr{N}}"'] \arrow[dr, "\pi"] & \\ \mathscr{N} & & \mathscr{B}. \end{tikzcd}$$

We see here that

$$\begin{aligned} \pi^{-1}(\mathfrak{b}) &= \left\{ x \in \mathfrak{b} \mid x \text{ is nilpotent} \right\} \\ &= \{ x \in \mathfrak{b} \mid \bar{x} = 0 \in \mathfrak{b}/[\mathfrak{b}, \mathfrak{b}] \} \end{aligned}$$

$$= [\mathfrak{b}, \mathfrak{b}] = \mathfrak{n}.$$

Note in fact that the condition of being nilpotent is equivalent to $P(x) = 0$ for all $P \in \mathscr{O}(\mathfrak{g})^G$. This identifies $\tilde{\mathscr{N}}$ as a G-equivariant vector bundle over $\mathscr{B}$ with fiber $\mathfrak{n}$, and in particular that

$$\tilde{\mathscr{N}} = G \times^B \mathfrak{n}.$$

Under this presentation, we see that $\mu(g, x) = g \cdot x$.

Theorem 3.18. *There is a natural identification $\tilde{\mathscr{N}} \cong T^*\mathscr{B}$.*

Proof. Note that if $\mathfrak{b} \in \mathscr{B}$, we have $T_{\mathfrak{b}}\mathscr{B} = \mathfrak{g}/\mathfrak{b}$, so the cotangent space $T_{\mathfrak{b}}\mathscr{B} = \mathfrak{b}^\perp \subset \mathfrak{g}^*$. Using the Killing form, we can identify $\mathfrak{g}^* \cong \mathfrak{g}$, and under this identification, $\mathfrak{b}^\perp = \mathfrak{n}$. Therefore, we have a canonical isomorphism $T^*\mathscr{B} \cong G \times^B \mathfrak{b}^\perp \cong G \times^B \mathfrak{n} \cong \tilde{\mathscr{N}}$. □

Later, we will see that $\mu_{\mathscr{N}}$ is the moment map for the G-action on $T^*\mathscr{B}$ induced by the G-action on $\mathscr{B}$. Let $\mathscr{O}(\mathfrak{g})^G_+ \subset \mathscr{O}(\mathfrak{g})^G$ be the functions which vanish at the origin.

Proposition 3.19. *$x \in \mathscr{N}$ if and only if $P(x) = 0$ for all $P \in \mathscr{O}(\mathfrak{g})^G_+$.*

Corollary 3.20. *The nilpotent cone $\mathscr{N}$ is an irreducible variety of dimension $2 \dim \mathfrak{n}$.*

Proof. The map $\mu_{\mathscr{N}}$ is proper and surjective, and $\tilde{\mathscr{N}} \cong T^*\mathscr{B}$ is irreducible. We also see that $\dim \mathscr{N} \leq \dim \tilde{\mathscr{N}} = 2 \dim \mathfrak{n}$. On the other hand, we know $\mathscr{O}(\mathfrak{g})^G_+ = \mathscr{O}(\mathfrak{H})^{\mathbb{W}}_+ = \langle f_1, \dots, f_r \rangle$, and the morphism $\chi \colon \mathfrak{g} \to \mathfrak{H} /\!/ \mathbb{W}$ expresses $\mathscr{N} = \chi^{-1}(0)$ and shows that $\dim \mathscr{N} \geq \dim \mathfrak{g} - r = 2 \dim \mathfrak{n}$. □

Proposition 3.21. *There are finitely many G-orbits in $\mathscr{N}$.*

Idea of proof. This is easy to see for $\mathfrak{g} = \mathfrak{gl}_n$ via the Jordan normal form. In the general case, we choose a faithful representation $\mathfrak{g} \hookrightarrow \mathfrak{gl}_n$, which embeds nilpotent cones by preservation of Jordan decomposition. The following lemma finishes the proof using the fact that $\mathfrak{g}$ is semisimple (so it has a $\mathfrak{g}$-stable complement in $\mathfrak{gl}_n$). □

Lemma 3.22. *Let $\mathfrak{g} \supset \mathfrak{a}$ such that $\mathfrak{a}$ has an $\mathfrak{a}$-stable complement $W \subset \mathfrak{g}$. Let G and A be connected algebraic groups with Lie algebras $\mathfrak{g}$ and $\mathfrak{a}$. For all G-orbits Ω in $\mathfrak{g}$, any irreducible component of $\Omega \cap \mathfrak{a}$ is a single A-orbit.*

Proof. Let $Z \subset \Omega \cap \mathfrak{a}$ be any irreducible component. For any $x \in Z$, we have

$$T_x Z \subset T_x \Omega = [x, \mathfrak{g}] = [x, \mathfrak{a}] + [x, W].$$

This implies that $T_x Z = [x, \mathfrak{a}] = T_x(A \cdot x)$, so $A \cdot x$ is open and dense in Z. This implies that for any $x_1, x_2 \in Z$, $A \cdot x_1 \cap A \cdot x_2 \neq \varnothing$, so the two orbits coincide. □

Corollary 3.23. *The nilpotent cone $\mathscr{N}$ contains an open dense orbit, which is exactly*

$$\mathscr{N}^{\mathrm{reg}} = \mathscr{N} \cap \mathfrak{g}^{\mathrm{reg}}.$$

Proof. For all $x \in \mathscr{N}^{\mathrm{reg}}$, we want to check that $\dim G \cdot x = \dim \mathscr{N}$. This follows from the fact that

$$\begin{aligned} \dim G \cdot x &= \dim G - \dim Z_G(x) \\ &= \dim G - r \\ &= 2 \dim \mathfrak{n}. \end{aligned}$$

□

Example 3.24. In the case of $\mathfrak{sl}_2$, we see that

$$\begin{aligned}\mathscr{N} &= \left\{\det\begin{bmatrix} x & y \\ z & -x \end{bmatrix} = 0\right\} \\ &= \left\{(x,y,z) \mid x^2 + yz = 0\right\}.\end{aligned}$$

See Figure 9 for a visualization of this variety. We can interpret $\tilde{\mathscr{N}} = T^*\mathbb{P}^1$ as

$$\tilde{\mathscr{N}} = \{(\xi, \ell) \mid \ell \in \ker \xi\}$$

and therefore anything in the open part is similar to $\left[\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right]$ and clearly has a one-dimensional kernel, while the origin has a two-dimensional kernel and thus its fiber is all of $\mathbb{P}^1$.

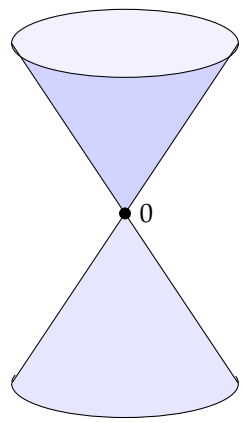


FIGURE 9. Nilpotent cone of $\mathfrak{sl}_2$

Now fix a $B \supset T$. This gives an action of T on $\mathfrak{n}/[\mathfrak{n},\mathfrak{n}] \cong \bigoplus_{\alpha_i \in \Pi} \mathfrak{g}_{\alpha_i}$. Choose a basis $e_i \in \mathfrak{g}_{\alpha_i}$ for $1 \le i \le r$. Define the subset

$$\mathfrak{n}^r = \left\{ x \in \mathfrak{n} \;\middle|\; \bar{x} = \sum_{i=1}^r a_i \bar{e}_i, a_i \neq 0 \text{ for all } i \right\}.$$

Proposition 3.25. *The set $\mathfrak{n}^r$ is a single B-orbit consisting of regular nilpotent elements.*

Proof. Note that $\mathfrak{n}^r$ is open and dense in $\mathfrak{n}$ (here, $\mathscr{N}^{\mathrm{reg}} \cap \mathfrak{n}$ must be open and dense in $\mathfrak{n}$), so $\mathfrak{n}^r$ must contain a regular element x. The image of $\mathfrak{n}^r$ in $\mathfrak{n}/[\mathfrak{n},\mathfrak{n}]$ is a single T-orbit, so it remains to show that $x + [\mathfrak{n},\mathfrak{n}]$ is a single U-orbit. On the level of tangent spaces, we see that

$$T_x(U \cdot x) = [\mathfrak{n}, x] \subset [\mathfrak{n},\mathfrak{n}]$$

and that this holds on all of $x + [\mathfrak{n},\mathfrak{n}]$. This tangent space has dimension

$$\dim U - \dim Z_U(x) \ge n - r,$$

and therefore $U \cdot x \subset x + [\mathfrak{n},\mathfrak{n}]$ is open and dense. However, the fact that U is unipotent and $x + [\mathfrak{n},\mathfrak{n}]$ is affine implies that it is also closed, so $U \cdot x = x + [\mathfrak{n},\mathfrak{n}]$. □

Proposition 3.26. *Every regular nilpotent element is contained in a single Borel.*

In particular, this means that the morphism $\mu\colon \tilde{\mathscr{N}} \to \mathscr{N}$ is an isomorphism over the regular locus.

Proof. Fix a regular nilpotent element

$$e = \sum_{i=1}^{r} e_i,$$

where each $0 \neq e_i \in \mathfrak{g}_{\alpha_i}$. Because this consists entirely of elements from the simple root spaces, there exists $h \in \mathfrak{h}$ such that $[h, e] = 2e$. This h is determined uniquely by the condition $\alpha_i(h) = 2$ for every simple root $\alpha_i \in \Pi$. This implies that for all positive roots $\alpha \in \Phi^+$, $\alpha(h) \neq 0$ and thus $h \in h^{\mathrm{reg}} \cap \mathbb{X}_\bullet(T)$.

Note that this h determines a one-parameter subgroup $i\colon \mathbb{C}^\times \to T$. The adjoint action is given by $\mathrm{Ad}_t(e) = t^2 e$ and lifts to an action of $\mathbb{C}^\times$ on $\mu^{-1}(e)$. For all $\mathfrak{b} \in \mu^{-1}(e)$, we see that

$$\mathrm{Ad}_t\, \mathfrak{b} \ni \mathrm{Ad}_t\, e = t^2 e,$$

and because $\mu^{-1}(e)$ is finite, the action is trivial. In particular, this implies that $\mathfrak{b}$ is a fixed point of the action of $\mathbb{C}^\times$ on $\mathscr{B}$ induced by i. This happens if and only if $h \in \mathfrak{b}$, and because h is regular, we see that $\mathfrak{h} \subset \mathfrak{b}$. The set of Borels satisfying this form a W-torsor, but any $\mathfrak{b}$ containing e must contain all simple root spaces $\mathfrak{g}_{\alpha_i}$, and there exists a unique Borel satisfying this condition. □

In particular, we see that $\tilde{\mathscr{N}} \to \mathscr{N}$ is a resolution of singularities, called the *Springer resolution*.

Remark 3.27. If $x \in \mathscr{N}^{\mathrm{reg}}$, then $\mu^{-1}(x)$ is a singleton. On the other hand, if $x \in \mathfrak{g}^{\mathrm{rs}}$, then $\mu^{-1}(x)$ is a $\mathbb{W}$-torsor. In general if $x \in \mathfrak{g}^{\mathrm{reg}}$, then $\mu^{-1}(x)$ is finite. More generally, the map $\mathfrak{g} \to \mathfrak{h} /\!/ W$ sends $x \mapsto [x_s]$ (after conjugating x to be in $\mathfrak{h}$). In particular, the set $\mu^{-1}(x)$ can be identified with the set $\mathbb{W}/\mathrm{Stab}_{\mathbb{W}}(x)$.

3.3. **Jacobson-Morozov theorem.** The idea is that for every nonzero $e \in \mathscr{N}$, we can find an $\mathfrak{sl}_2$-triple (e, h, f).

Theorem 3.28. *Let $0 \neq e \in \mathscr{N}$. Then there exist $h, f \in \mathfrak{g}$ such that h is semisimple, f is nilpotent, $[h, e] = 2e$, $[h, f] = -2f$, and $[e, f] = h$.*

Note that in $\mathfrak{sl}_2$, the matrices

$$h = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \qquad e = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \qquad f = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$$

satisfy these relations. For a general $\mathfrak{g}$, these give an embedding $\mathfrak{sl}_2 \hookrightarrow \mathfrak{g}$ sending the standard basis to (e, h, f). Also, in the general case, $e + \ker \mathrm{ad}_f = e + \mathfrak{g}^f$ is a transverse slice to $G \cdot e$, and if e is regular, it is called the *principal slice* or *Kostant slice* and satisfies $e + \mathfrak{g}^f = \mathfrak{h} /\!/ W$.

Corollary 3.29. *For all $e \in \mathscr{N}$, there exists a rational group homomorphism $\mathrm{SL}_2 \to G$ sending $\left[\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right]$ to e.*

Proof of Theorem 3.28. Recall the Killing form

$$(-,-)\colon \mathfrak{g} \times \mathfrak{g} \to \mathbb{C} \qquad (x, y) \mapsto \mathrm{tr}(\mathrm{ad}_x\, \mathrm{ad}_y).$$

The fact that this is G-invariant means that

$$([x, y], z) = (x, [y, z]).$$

To simplify things, we will write $G^x = Z_G(x)$ and $\mathfrak{g}^x = \mathfrak{Z}_{\mathfrak{g}}(x) = \ker \operatorname{ad}_x$ for $x \in \mathfrak{g}$.

Lemma 3.30. *Assume that $e \in \mathscr{N}$ and $\mathfrak{g}^e$ consists entirely of nilpotent elements. Then $(e, \mathfrak{g}^e) = 0$.*

Proof. We need to show that for any $x \in \mathfrak{g}^e$, the trace $\operatorname{tr}(\operatorname{ad}_e \operatorname{ad}_x) = 0$. However, both ad_e and ad_x are nilpotent, and they commute, so their product is also nilpotent. □

Our goal now is to reduce to the case where $\mathfrak{g}^e \subset \mathscr{N}$ so that we can induct on the rank of $\mathfrak{g}$. Assume that if the rank is strictly less than r, then the theorem holds. For all $e \in \mathscr{N}$, we consider the set $\mathfrak{g}^e$. If $\mathfrak{g}^e \not\subset \mathscr{N}$, then there exists $x \in \mathfrak{g}^e$ which is not nilpotent. Writing $x = x_s + x_n$, we see that $[e, x] = 0$ implies that both $[e, x_s] = 0$ and $[e, x_n] = 0$, so $e \in \mathfrak{g}^{x_s} \cap \mathfrak{g}^{x_n}$. Because x_s is semisimple and nonzero, $\mathfrak{g}^{x_s} \subset \mathfrak{g}$ is a proper Levi subalgebra and is hence reductive. Because e is nilpotent, we have

$$e \in [\mathfrak{g}^{x_s}, \mathfrak{g}^{x_s}] = \mathfrak{g}^{x_s, \mathrm{der}},$$

which is semisimple and has strictly smaller rank than $\mathfrak{g}$.

In the case where $\mathfrak{g}^e \subset \mathscr{N}$, we use the ad_e-invariance of the Killing form to obtain

$$([e, x], y) = -(x, [e, y])$$

and therefore that $\operatorname{Im} \operatorname{ad}_e \subset (\ker \operatorname{ad}_e)^\perp = (\mathfrak{g}^e)^\perp$. Because the Killing form is nondegenerate, the two are actually equal. This implies that $e \in \operatorname{Im} \operatorname{ad}_e$ and thus that there exists $h \in \mathfrak{g}$ such that $[h, e] = 2e$. Decomposing $h = s + n$ into a semisimple and nilpotent part, we see that $[s, e] = 2e$. Therefore, we can replace h by s.

Our final goal is to find f. Consider the weight decomposition of $\operatorname{ad}_h \colon \mathfrak{g} \to \mathfrak{g}$. This decomposes

$$\mathfrak{g} = \bigoplus_{k \in \mathbb{C}} \mathfrak{g}_k.$$

Here, note that $h \in \mathfrak{g}_0$ and $e \in \mathfrak{g}_2$, and because $[h, e] = 2e$, we see

$$\operatorname{ad}_e \colon \mathfrak{g}_k \to \mathfrak{g}_{k+2}.$$

Here, the calculation is

$$\begin{aligned} [h, [e, x]] &= [[h, e], x] + [e, [h, x]] \\ &= 2[e, x] + k[e, x] \\ &= (k + 2)[e, x]. \end{aligned}$$

We want to find $f \in \mathfrak{g}_{-2}$ such that $[e, f] = h$, or equivalently, show that $h \in \operatorname{Im} \operatorname{ad}_e = (\mathfrak{g}^e)^\perp$. Because $[h, e] = 2e$ and $[h, \mathfrak{g}^e] \subset \mathfrak{g}^e$, we see that the subspace $\mathbb{C}h + \mathfrak{g}^e$ is a solvable subalgebra in $\mathfrak{g}$. By the theorem of Lie and Engel, there exists a basis of $\mathfrak{g}$ such that ad_h is upper-triangular and for any $x \in \mathfrak{g}^e$, ad_x is strictly upper-triangular. This implies that

$$\operatorname{tr}(\operatorname{ad}_h \operatorname{ad}_x) = 0$$

for any $x \in \mathfrak{g}^e$ because $\operatorname{ad}_h \operatorname{ad}_x$ is strictly upper-triangular. This implies that $h \in (\mathfrak{g}^e)^\perp = \operatorname{Im} \operatorname{ad}_e$, so there exists f such that $[e, f] = h$. Because $h \in \mathfrak{g}_0$, we can choose $f \in \mathfrak{g}_{-2}$. □

Proposition 3.31. *All $\mathfrak{sl}_2$-triples containing $e \in \mathscr{N}$ are conjugate under $G^e := Z_G(e)$.*

For a fixed triple (e, h, f), we can consider the induced representation of $\mathfrak{sl}_2$ from any representation of $\mathfrak{g}$. Recall that representations of $\mathfrak{sl}_2$ are classified by their highest weight, which is a positive integer n. See Figure 10 for a visualization. For non-irreducible representations, we can draw a diagram with all of the irreducible components.

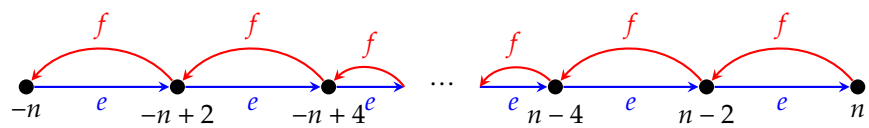


FIGURE 10. Irreducible $\mathfrak{sl}_2$-representation V_n with highest weight n.

Specifying to $V = \mathfrak{g}$, we see that $\mathfrak{g} = \mathfrak{g}^f \oplus \mathfrak{I}\,\mathrm{ad}_e$. If we decompose

$$\mathfrak{g} = \bigoplus_{k \in \mathbb{Z}} \mathfrak{g}_k,$$

note that $[g_k, \mathfrak{g}_\ell] \subset \mathfrak{g}_{k+\ell}$.

Remark 3.32. Note that $\dim \mathfrak{g}^e = \dim \mathfrak{g}^f \geq \dim \mathfrak{g}^h = \dim \mathfrak{g}_0$ and equality holds if and only if all weights are even.

If e is regular, then $\dim \mathfrak{g}^e = r$, and therefore f is regular and $\dim \mathfrak{g}^h = \dim \mathfrak{g}^e = r$, and therefore e, f, h are all regular. In this case, we have a *principal* $\mathfrak{sl}_2$*-triple*. Note here that all weights are even. Also, note that

$$\mathfrak{g}^e \subseteq \bigoplus_{k \geq 0} \mathfrak{g}_k$$

but is not necessarily contained in the strictly positive part. By definition, set

$$\mathfrak{u} = \mathfrak{g}^e \cap \bigoplus_{k>0} \mathfrak{g}_k = \mathfrak{g}^e \cap [\mathfrak{g}, e] = \ker \mathrm{ad}_e \cap \mathrm{Im}\, \mathrm{ad}_e\,.$$

By definition, we have $V_{\mathrm{triv}} = \mathfrak{Z}(e, f, h) \subset \mathfrak{g}$, and if there is no trivial part under $\mathfrak{sl}_2$, then $\mathfrak{g}^e$ is contained in the strictly positive part. Now lift $\mathfrak{u}$ to a unipotent subgroup $U \subset G^e$. Note that G^e may be disconnected, but U is connected.

Lemma 3.33. *The affine space $h + \mathfrak{u}$ is a single U-orbit.*

Proof. First, note that if $x \in \mathfrak{u}$, then $[h + x, \mathfrak{u}] \subset \mathfrak{u}$ because $\mathfrak{u}$ consists of weight spaces. In fact, they are actually equal because $[h, \mathfrak{u}] = \mathfrak{u}$. By a standard argument, the orbit $U \cdot h \subset h + \mathfrak{u}$ is open, but U is unipotent, so it is also closed. □

Proof of Proposition 3.31. Fix (e, h, f) and (e, h', f'). Note that $[h' - h, e] = 0$, so $h' - h \in \mathfrak{g}^e$. In particular, we see that $h' - h \in \mathfrak{g}^e \cap [\mathfrak{g}, e]$ because we also have $[e, f' - f] = h' - h$. Therefore, $h' \in h + \mathfrak{u} = U \cdot h$, so there exists $u \in U$ such that $u \cdot h = h'$.

Now we have $u \cdot (e, h, f) = (e, h', u \cdot f)$. Note that

$$[e, uf] = h' = [e, f'],$$

and therefore $[e, uf - f'] = 0$. This implies that $uf - f' \in \mathfrak{g}^e$, but $uf - f' \in \mathfrak{g}_{-2}$ while $\mathfrak{g}^e$ is concentrated in nonnegative weights, so $uf = f'$. Therefore, $u \cdot (e, h, f) = (e, h', f')$. □

Remark 3.34. We have in fact also shown that if we fix e and h, then f is unique.

We will now consider the structure of G^e.

Proposition 3.35. *The unipotent subgroup U is the unipotent radical of G^e and $H := Z_G(e,h,f)$ is the maximal reductive subgroup of G^e.*

Example 3.36. If e is regular, then so are h and f. When we described $\mathscr{N}^{\mathrm{reg}}$, recall that $\mathfrak{n}^{\mathrm{reg}}$ consisted of all elements which are nonzero on all simple root spaces. If we let

$$e = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ & 0 & 1 & \cdots & 0 \\ & & \ddots & \ddots & \vdots \\ & & & 0 & 1 \\ & & & & 0 \end{bmatrix},$$

thhen h is actually unique is given by

$$h = \begin{bmatrix} n-1 & & & & \\ & n-3 & & & \\ & & \ddots & & \\ & & & 3-n & \\ & & & & 1-n \end{bmatrix}$$

while f is lower-triangular with nonzero entries on the subdiagonal. In this case, we have $Z_G(e,h,f) = Z(G)$. For all $g \in Z_G(e,h,f)$, we see that $g \in Z_G(h) = T$, and so under some choice of Borel $\mathfrak{b}$ containing $h \in \mathfrak{h}^{\mathrm{reg}}$, we see that

$$g \cdot e_\alpha = \alpha(g) e_\alpha$$

and therefore if $g \in Z_G(e)$, this is equivalent to $\alpha(g) = 1$ for all simple roots α, which is equivalent to $g \in Z(G)$.

Because e is regular, we see that $\dim G^e = r$. Using the proposition, we see that $\dim U = r$, and therefore that $\mathfrak{g}^e = \mathfrak{u}$. Here, we see that

$$\begin{aligned} h &= 2\rho^\vee \\ &= 2\sum_{i \in \Pi} \omega_i^\vee \\ &= \sum_{\alpha \in \Phi^+} \alpha^\vee. \end{aligned}$$

Looking at the decomposition of $\mathfrak{g}$ with respect to this h, then in fact

$$\mathfrak{g} = \bigoplus_{k \in 2\mathbb{Z}} \mathfrak{g}_k,$$

or namely all weights are even. There is no trivial $\mathfrak{sl}_2$ representation appearing as a summand of $\mathfrak{g}$ because $Z_G(e,h,f)$ is discrete.

Proof of Proposition 3.35. Our goal is to show that $G^e = HU$ and that $H \cap U$ is trivial. We already know that $U \lhd G^e$ because $\mathfrak{u}$ is an ideal in $\mathfrak{g}^e$.

We will use the fact that the centralizer of a reductive subgroup in G is always reductive. Because the image of SL_2 in G given by e,h,f is reductive, so is H. We also know that $H \cap U \lhd H$, and because H is reductive, the intersection is trivial.

Finally, we need to prove that $G^e = HU$. For any $g \in G^e$, recall that

$$g \cdot (e,h,f) = (e,h',f') = u \cdot (e,h,f)$$

for some $u \in U$. Therefore, we see that $u^{-1}(g) \in Z_G(e,h,f)$ and therefore $g \in UH = HU$ (here, we use normality of U). □

3.4. **Regular $\mathfrak{sl}_2$-triples.** Let $e \in \mathscr{N}^{\mathrm{reg}}$ and consider the Kostant slice $e + \mathfrak{g}^f$. Recall that the map $\chi\colon \mathfrak{g} \to \mathfrak{h} /\!/ W$ restricts to an isomorphism

$$\chi|_{e+\mathfrak{g}^f}\colon e + \mathfrak{g}^f \xrightarrow{\sim} \mathfrak{h} /\!/ W.$$

Note that $\mathscr{O}(\mathfrak{g})^G \simeq \mathscr{O}(\mathfrak{h})^W$ is an isomorphism of graded rings.

Theorem 3.37 (Chevalley-Shephard-Todd)**.** *We have $\mathscr{O}(\mathfrak{h})^W \simeq \mathbb{C}[P_1, \dots, P_r]$, where $P_1, \dots, P_r$ are homogeneous elements of degrees $d_1 \le \cdots \le d_r$ which are independent of the choice of generators. These $d_1, \dots, d_r$ are called the exponents of $\mathfrak{g}$.*

Theorem 3.38 (Kostant)**.** *Let (e,h,f) be a regular $\mathfrak{sl}_2$-triple. Recall that the decomposition*

$$\mathfrak{g} = \bigoplus_j V_j$$

into irreducible $\mathfrak{sl}_2$-representations V_j satisfies $\dim V_j = 2\mu_j + 1$, where $\mu_j > 0$. Order V_j such that $1 = \mu_1 \le \cdots \le \mu_k$. Then $k = r$ and $d_j = \mu_j + 1$.

Proof. Consider the decomposition

$$\mathfrak{g} = \mathfrak{g}^f + [\mathfrak{g}, e].$$

For each V_j, the lowest weight is $-2\mu_j$. Because $\mathfrak{g}^f \cap V_j$ is the lowest weight space of V_j, we can let the set of $0 \neq \xi_j \in \mathfrak{g}^f \cap V_j$ be a basis of $\mathfrak{g}^f$. Define a map

$$\psi\colon G \times \mathbb{C}^r \to \mathfrak{g} \qquad (g, a_1, \dots, a_r) \mapsto g \cdot \left(e + \sum_{j=1}^r a_j \xi_j\right).$$

We will show that $\mathrm{d}\psi$ is surjective everywhere, or namely that ψ is a submersion. Because this morphism is G-equivariant, it suffices to check when $g = 1$. Let $x = (1, a_1, \dots, a_r)$. Then

$$\mathrm{d}_x\psi\colon \mathfrak{g} \times \mathbb{C}^r \to \mathfrak{g} \qquad (\eta, b_1, \dots, b_r) \mapsto \left[\eta, e + \sum_j a_j \xi_j\right] + \sum_j b_j \xi_j.$$

Our goal is to prove that this is surjective. We compute

$$\begin{aligned}
\mathrm{Im}(\mathrm{d}_x\psi) &= [\mathfrak{g}, e + \sum_j a_j \xi_j] + \sum_j b_j \xi_j \\
&= \left[\mathfrak{g}, e + \sum_j a_j \xi_j\right] + \mathfrak{g}^f \\
&= [\mathfrak{g}, e] + \mathfrak{g}^f \\
&= \mathfrak{g}.
\end{aligned}$$

□

Part 2. Universal centralizers